

**İZMİR UNIVERSITY OF ECONOMICS**

**Department of Aerospace Engineering**

*FENG 498: Engineering Capstone*

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**Comparative Energy Consumption and CO<sub>2</sub> Emissions Analysis of  
Conventional, Hybrid and Electromagnetic Propulsion Systems for UAV  
Flight Stages: A Case Study on the Bayraktar TB2 MALE UAV**

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## 1. ABSTRACT

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This report presents a comparative energy efficiency and CO<sub>2</sub> emissions analysis of three propulsion configurations for the Bayraktar TB2 Medium Altitude Long Endurance (MALE) Unmanned Aerial Vehicle (UAV): a conventional Internal Combustion Engine (ICE), a hybrid-electric system, and a purely electromagnetic drive. The study is motivated by the growing operational deployment of MALE UAVs and the need to quantify the environmental and energy trade-offs between propulsion technologies under a realistic ISR mission profile.

For the ICE configuration—the primary focus of this work—a reduced-order MATLAB mission simulation was constructed following the segmented flight-performance methodology of Masood and Wei [1]. The aircraft model was calibrated against the published benchmark, yielding a BSFC scaling factor of 1.2493 that was then transferred to a TB2-class ISR mission (Takeoff → Climb → Transit Out → Loiter ISR 4 h → Transit Home → Descent → Landing). The optimised mission produced a total fuel consumption of 45.97 kg, a path length of 782.59 km, a mission time of 6.20 h, and an estimated tailpipe CO<sub>2</sub> of 145.25 kg. An atmosphere sensitivity study confirmed that a fixed-atmosphere simplification introduces a maximum output deviation of 0.06%, justifying its use for the subsequent propulsion comparisons.

A Python-based weather decision application was developed alongside the MATLAB simulation. It fetches live Open-Meteo forecast and air quality data for a user-specified location and time, evaluates ten weather constraints derived from the TB2 operational envelope, and returns a MISSION APPROVED / CAUTION / MISSION REJECTED decision together with optimised flight-parameter recommendations.

For the hybrid-electric configuration, an actively managed parallel propulsion framework was modeled using a 120 kg semi-solid-state battery pack providing a 42.0 kWh capacity alongside a supplementary 100 hp electric motor. Under a rule-based 50/50 power-split strategy executed during the transit and loiter surveillance legs, the model successfully restricted the internal combustion engine to its lowest brake specific fuel consumption (BSFC) operating regime. Dynamic time-domain script integration demonstrates that this hybrid approach achieves a major fuel and tailpipe CO<sub>2</sub> emissions reduction of 39.58% compared to the calibrated conventional baseline, bringing total fuel burn down to 27.77 kg while maintaining a safe 10.00% battery state-of-charge margin.

For the third architecture, an electrically assisted (EM-assisted) propulsion framework was simulated, integrating high-frequency energy storage elements to absorb transient maneuvering loads alongside the primary hydrocarbon core. Time-domain simulation curves indicate that while this dual-source configuration provides key torque enhancements during short dynamic segments like takeoff and rapid climb, it leaves the internal combustion engine dominant during the prolonged, energy-dense loiter window. Consequently, the EM-assisted setup yields a total source energy of approximately 2020 MJ, a tailpipe CO<sub>2</sub> output of 145 kg, and a mission-level efficiency of 0.42 km/MJ, remaining

near the conventional baseline due to the unoptimized scaling of its electrical endurance assist.

The unified comparison framework—applying the same mission profile and KPI set to all three configurations—provides a consistent basis for evaluating the trade-offs between energy efficiency, operational range, and environmental impact in next-generation UAV propulsion.

## 2. INTRODUCTION

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### 2.1 Problem Statement

Medium Altitude Long Endurance UAVs have become central to modern surveillance, reconnaissance, and precision engagement operations. The Bayraktar TB2, designed and manufactured by Baykar Makina, represents one of the most operationally deployed platforms in this class, with a Maximum Take-Off Weight (MTOW) of 700 kg, a 100 hp internal combustion engine, and a documented endurance exceeding 20 hours [2]. Despite its operational maturity, the propulsion system of the TB2 remains dependent on hydrocarbon fuel, with direct implications for energy consumption, logistical supply chains, and greenhouse gas emissions.

The engineering problem addressed by this project is the quantitative comparison of energy efficiency and CO<sub>2</sub> emissions across three propulsion architectures—ICE, hybrid-electric, and fully electromagnetic—applied to the same TB2-class platform and mission profile. Accurate modelling of each configuration is required before a meaningful propulsion trade-off can be made, and each model must be validated or calibrated against published flight-performance data.

### 2.2 Motivation

The motivation for this study is threefold. First, fuel dependency increases logistical complexity and vulnerability in forward-deployed UAV operations. Second, few studies provide a unified, mission-level energy and emissions comparison across ICE, hybrid, and electric configurations for the same MALE-class airframe [3, 4]. Third, ICAO's Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) is driving requirements for lower-carbon propulsion across all aircraft categories. The TB2 platform was selected because it provides a well-documented public specification set [2], has been studied in the aeronautical literature [1, 3], and represents a realistic near-term candidate for propulsion modernisation.

Conventional internal combustion architectures for tactical MALE UAVs must be mechanically oversized to meet peak takeoff and climb requirements, leaving them to operate under highly inefficient part-load conditions during prolonged loiter windows. Integrating a parallel hybrid-electric system addresses this bottleneck by allowing the primary engine to lock continuously into its most efficient, narrow fuel-burn band while the electric motor absorbs transient load variations. Additionally, this configuration provides a tactical mechanism to

completely deactivate the engine during the final approach, enabling a silent, zero-emission landing mode that dramatically minimizes acoustic and thermal visibility.

Conventional MALE UAV powerplants must be structurally oversized to handle severe, short-duration mechanical surges during takeoff and rapid maneuvering, which significantly penalizes airframe weight and efficiency. Transitioning toward an electrically assisted (EM-assisted) propulsion model utilizing a dual-source battery-supercapacitor matrix solves this challenge by isolating fast-transient power spikes within high-rate electrical buffers. Investigating this configuration is highly motivated by the need to protect primary energy systems from high-frequency thermal stress and current ripples, thereby establishing the precise operational trade-offs of localized electrical transient support across tactical ISR mission windows.

## 2.3 Sustainable Development Goals (SDGs)

This project aligns with three United Nations Sustainable Development Goals:

- **SDG 7 — Affordable and Clean Energy:** Substituting conventional ICE propulsion with hybrid or electromagnetic alternatives directly reduces dependence on fossil fuels in aerospace applications. Quantifying the energy difference between configurations provides an evidence base for cleaner propulsion feasibility.
- **SDG 9 — Industry, Innovation and Infrastructure:** The development of a calibrated, modular MATLAB simulation framework and a weather-aware mission decision application represents a methodological innovation in UAV mission planning tools, transferable to other fixed-wing UAV platforms.
- **SDG 13 — Climate Action:** The CO<sub>2</sub> estimate generated by the ICE model (145.25 kg per mission) provides a quantified baseline against which hybrid and electromagnetic configurations can be evaluated, contributing to aviation's decarbonisation pathway.

## 3. LITERATURE REVIEW

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### 3.1 Academic and Technical Publications

#### 3.1.1 Primary Benchmark: Masood and Wei (2012)

The primary calibration reference is Masood and Wei [1], which presents a comprehensive fixed-wing UAV mission analysis: a nine-segment profile with a 540 kg UAV, 110 kg fuel, 9 m<sup>2</sup> wing area, and 100 hp Rotax 914UL engine. Stage-by-stage fuel consumption, range, altitude, and total mission metrics are provided in tabular form, making it uniquely suitable as a calibration anchor. The calibrated model achieved a fuel-consumption error of only -0.83% against the published 101 kg total.

#### 3.1.2 TB2 Aerodynamic and Performance Studies

Panagiotou et al. [3] present a detailed aerodynamic design study for a comparable MALE UAV, supporting the use of a parabolic drag polar for preliminary-level TB2 performance analysis. Rajendran [5] provides methodological justification for reduced-order point-mass simulation. Octavianty et al. [6] and Diogo and Fernandes [7] address endurance optimisation and propulsion selection for tactical UAV classes, reinforcing the case for a systematic KPI-based propulsion comparison. Gotten et al. [4] examine energy management for hybrid UAV systems and provide the comparative framework that motivates the three-configuration study.

A core methodological foundation for the hybrid-electric flight mechanics in this study is derived from the parametric work of Benmoussa and Gamboa [32]. The authors established a comprehensive longitudinal flight simulation framework to evaluate how variations in localized flight control parameters directly dictate the energy efficiency and transient behavior of hybrid-electric UAVs. Rather than relying on classical, static energy-matching approximations, their methodology models the aircraft's states continuously across highly transient, multi-segmented mission profiles, with a specific focus on the demanding deceleration transition phases.

Their findings demonstrate that when transitioning between flight states—such as moving from a fixed-wing high-speed cruise down to a localized vertical-landing or loiter state—the pitch tracking parameters ( $\delta$  and  $T\delta$ ) govern the system's efficiency. Poorly tuned or "average" baseline control vectors induce severe pitch oscillations ( $\delta$ ), which delay velocity stabilization, force the electric propulsion motors to draw sustained peak currents from the onboard battery pack, and drastically compound the overall energy consumption. This study directly adapts Benmoussa and Gamboa's continuous two-dimensional x-z flight dynamic scripts to isolate how optimized versus unoptimized control tuning influences the battery state-of-charge (SoC) and fuel-burn profiles of a TB2-class tactical UAV during transitional segments.

Academic literature regarding aviation electrification emphasizes that the mission-level viability of battery-powered flight remains strictly constrained by weight restrictions, battery thermal management overhead, and altitude-dependent power requirements [36, 37]. To mitigate these boundaries, modern research focuses on utilizing dual-source Hybrid Energy Storage Systems (HESS)—such as a battery-supercapacitor matrix—to isolate high-frequency power surges and protect primary energy architectures from rapid degradation [38]. These foundational studies support evaluating the EM configuration as an electrically assisted system, where localized transient support must be analyzed alongside the aircraft's dominant endurance requirements.

### 3.1.3 Atmospheric Modelling and Sensitivity Analysis

The International Standard Atmosphere [8], implemented in MATLAB via `atmosisa` [9], provides altitude-dependent air properties. NASA Glenn educational material [10, 11] derives the lift and drag equations that connect atmospheric density to aerodynamic coefficients through  $q = \frac{1}{2}\rho V^2$ . A controlled one-factor sensitivity study comparing variable-atmosphere and fixed-atmosphere models was conducted following the experimental-design logic of the NIST/SEMATECH e-Handbook [12] and Saltelli et al. [13].

### 3.1.4 Weather-Aware UAV Decision Systems

The Open-Meteo Forecast API [14], WMO weather codes [15], and Open-Meteo Air Quality API [16] form the data foundation of the weather decision application. The FAA Pilot's Handbook of Aeronautical Knowledge [17] provides the decision-making framework: wind, precipitation, visibility, temperature extremes, and icing are each identified as flight-safety factors. FastAPI [18, 19] and Streamlit [20] were selected for rapid prototyping of the scientific decision tool.

## 3.2 Intellectual Property, Standards and Regulations

The publicly available Baykar specifications [2] are used without modification. Relevant standards and frameworks include:

- ICAO Annex 16, Volume II — Aircraft Engine Emissions.
- ICAO CORSIA — Carbon Offsetting and Reduction Scheme for International Aviation.
- NATO STANAG 4671 — UAV Airworthiness Requirements.
- FAA Advisory Circular AC 107-2 — Small Unmanned Aircraft Systems.
- MIL-STD-810H — Environmental Engineering Considerations and Laboratory Tests.
- RTCA DO-311A / ASTM F3363 — Onboard Energy Storage Safety.
- SAE AS50881 — Electrical Wiring, Insulation, and High-Voltage Safety.
- SAE AIR6198 — Aerospace High Voltage Direct Current Power System Factors.
- MIL-STD-704F — Aircraft Electric Power Characteristics.
- IEC 62391-2 — Fixed Electric Double-Layer Capacitors for Power Applications.
- RTCA DO-160G — Environmental Conditions and Test Procedures for Airborne Equipment.

## 4. DESIGN METHODOLOGY

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### 4.1 Design Limitations and Realistic Constraints

- **Technical:** Detailed TB2 aerodynamic polar, propeller map, and fuel-system data are unavailable; engineering assumptions (parabolic polar, constant  $\eta_{\text{prop}}$ , fixed BSFC) are calibrated against published data.
- **Economic:** Exclusively open-source tools (MATLAB, Python, Open-Meteo API) ensure reproducibility without commercial licence dependencies.
- **Environmental:** CO<sub>2</sub> estimates are confined to tailpipe emissions; lifecycle emissions are outside scope but noted as a limitation.
- **Safety:** All MISSION APPROVED / CAUTION / MISSION REJECTED thresholds are explicitly stated and traceable to FAA PHAK [17], UAV operational literature, and manufacturer envelope data.
- **Scope:** All three models are compared under identical mission profiles and atmospheric conditions.

## 4.2 Design Options

- **Option A — ICE Only (Baseline):** Rotax-class 100 hp ICE; highest specific power and longest demonstrated endurance; highest direct CO<sub>2</sub> and fuel-logistics dependency.
- **Option B — Hybrid-Electric:** Series or parallel ICE-battery hybrid; reduces fuel consumption in low-power phases; retains range for high-power phases.
- **Option C — Electromagnetic:** All-electric motor-propeller drive; eliminates tailpipe CO<sub>2</sub>; constrained by current battery energy density.

## 4.3 Suggested Design and Selection Rationale

The quantitative comparison across all three options is the main project output. The ICE model (this work) provides the 145.25 kg CO<sub>2</sub> per mission baseline.

1. ICE: highest endurance and simplest implementation; 145.25 kg CO<sub>2</sub> per mission.
2. Hybrid: expected fuel reduction during loiter and descent; 87.765 kg CO<sub>2</sub> per mission.
3. Electromagnetic: expected zero direct CO<sub>2</sub>; 125 kg CO<sub>2</sub> per mission.

## 4.4 Project Management

### 4.4.1 Work Packages, Timeline and Roles

| Work Package                            | Author 1<br>(ICE+App) | Author 2<br>(Hybrid) | Author 3 (EM) | Deadline |
|-----------------------------------------|-----------------------|----------------------|---------------|----------|
| Literature review & benchmark selection | R/A                   | R/A                  | R/A           | Week 2   |
| MATLAB ICE simulation                   | R/A                   | I                    | I             | Week 5   |
| MATLAB Hybrid simulation                | I                     | R/A                  | I             | Week 7   |
| MATLAB Electromagnetic simulation       | I                     | I                    | R/A           | Week 7   |
| Python weather decision app             | R/A                   | I                    | I             | Week 6   |
| Atmosphere sensitivity study            | R/A                   | I                    | I             | Week 5   |
| KPI comparison & cross-model tables     | R                     | R                    | R             | Week 9   |
| Report writing & integration            | R                     | R                    | R             | Week 11  |
| Final review & submission               | A                     | A                    | A             | Week 12  |

Table 1. RACI Matrix. R = Responsible, A = Accountable, C = Consulted, I = Informed.

### 4.4.2 Risk Analysis

| Risk | Likelihood | Impact | Mitigation |
|------|------------|--------|------------|
|------|------------|--------|------------|

| Risk                                     | Likelihood      | Impact | Mitigation                                    |
|------------------------------------------|-----------------|--------|-----------------------------------------------|
| MATLAB simulation diverges               | Low             | High   | Reduce $\Delta t$ ; add Euler step limiting   |
| TB2 aero data unavailable                | High (realised) | Medium | Parabolic polar + BSFC calibration            |
| Open-Meteo API unavailable               | Low             | Medium | Cache last response; offline mode             |
| Electromagnetic model range insufficient | Medium          | High   | Adjust loiter duration                        |
| Battery density assumptions outdated     | Medium          | High   | Conservative values + sensitivity study       |
| Co-author integration delays             | Medium          | Medium | Placeholder sections allow partial submission |

*Table 2. Risk Analysis Matrix.*

## 4.5 Technical Design Framework

### 4.5.1 Engineering Calculations — ICE Simulation Model

The ICE mission simulation implements a point-mass segmented flight-performance analysis. The governing equations are presented in standard aerospace notation, consistent with Anderson [21], Raymer [22], and McCormick [23].

International Standard Atmosphere, troposphere ( $h \leq 11\,000$  m) [8, 9]:

$$T(h) = T_0 - L \cdot h \quad (4.1)$$

$$P(h) = P_0 \cdot \frac{T(h)^{\frac{g}{R \cdot L}}}{T_0^{\frac{g}{R \cdot L}}} \quad (4.2)$$

$$\rho(h) = \frac{P(h)}{R \cdot T(h)} \quad (4.3)$$

$$a(h) = \sqrt{\gamma \cdot R \cdot T(h)} \quad (4.4)$$

where  $T_0 = 288.15$  K,  $P_0 = 101\,325$  Pa,  $L = 0.0065$  K/m,  $g = 9.81$  m/s<sup>2</sup>,  $R = 287.05$  J/(kg·K),  $\gamma = 1.4$ .

Dynamic pressure and aerodynamic forces [10, 11]:

$$q = \frac{1}{2} \rho \cdot V^2 \quad (4.5)$$

$$L = m \cdot g \cdot \cos(\gamma_{path}) \quad (4.6)$$

$$C_L = \frac{L}{q \cdot S} \quad (4.7)$$

$$C_D = C_{D0} + k \cdot C_L^2, \quad k = \frac{1}{\pi \cdot e \cdot AR} \quad (4.8)$$

$$D = q \cdot S \cdot C_D \quad (4.9)$$

Stall speed and operational V-speeds [17, 22]:

$$V_{stall} = \sqrt{\frac{2W}{\rho \cdot S \cdot C_{L,max}}} \quad (4.10)$$

$$V_{TO} = 1.20 \cdot V_{stall} \quad (4.11)$$

$$V_{app} = 1.30 \cdot V_{stall} \quad (4.12)$$

Thrust balance along the flight path (point-mass model) [23, 24]:

$$T_{req} = D + m \cdot a_{long} + m \cdot g \cdot \sin(\gamma_{path}) \quad (4.13)$$

$$P_{req} = \frac{T_{req} \cdot V}{\eta_{prop}} \quad (4.14)$$

$$P_{used} = \min(\max(P_{req}, P_{idle}), P_{max}) \quad (4.15)$$

Fuel consumption and mass depletion [1, 22]:

$$\dot{m}_{fuel} = BSFC \cdot \frac{P_{used} [kW]}{3600} [kg/s] \quad (4.16)$$

$$m_{fuel, k+1} = m_{fuel, k} + \dot{m}_{fuel}(k) \cdot \Delta t \quad (4.17)$$

$$m(k+1) = m_{dry} + (m_{fuel, 0} - m_{fuel, k+1}) \quad (4.18)$$

Fuel energy and CO<sub>2</sub> estimation [25]:

$$\dot{E}_{fuel} = \dot{m}_{fuel} \cdot LHV [W] \quad (4.19)$$

$$CO_2 [kg] = m_{fuel, used} \cdot 3.16 \quad (4.20)$$

Speed optimisation (transit: best range; loiter: best endurance) [21, 22]:

$$V_{transit} = \arg \min_V \frac{\dot{m}_{fuel}}{V} \quad (4.21)$$

$$V_{loiter} = \arg \min_V \dot{m}_{fuel} \quad (4.22)$$

BSFC calibration factor (Eq. 4.23) transfers the fuel model from the paper benchmark to the TB2-class mission:

$$k_{BSFC} = \frac{m_{fuel, published}}{m_{fuel, uncal}} = \frac{101.0}{80.848} = 1.2493 \quad (4.23)$$

#### 4.5.2 Engineering Calculations — Hybrid Propulsion Model

The hybrid propulsion architecture operates as an actively managed parallel configuration. In this arrangement, a supplementary electric motor (100 hp / 74.57 kW peak capacity) and a primary compression-ignition internal combustion engine (ICE, 100 hp baseline) are coupled to supply power to the propulsive shaft. Onboard energy is managed via a 120 kg semi-solid-state battery system possessing a specific energy density of 350 Wh/kg, yielding a total nominal capacity ( $E_{cap}$ ) of 42.0 kWh.

The calculation sequences used to evaluate the transient performance matrices across the distinct flight stages are derived below:

##### 1. Longitudinal Flight Kinematics and Multi-Phase Trajectory Balance

The dynamic equations of motion track the translation of the 700 kg tactical UAV platform within the vertical  $x$ - $z$  plane. At any discrete time index  $k$ , the localized flight path angle ( $\gamma$ ) is extracted from the instantaneous vertical climb rate ( $\dot{h}$ ) and forward airspeed ( $V$ ):

$$\gamma(k) = \arcsin \left( \frac{\dot{h}(k)}{V(k)} \right)$$

Resolving the translational force balance along the horizontal flight line and vertical altitude column establishes the net instantaneous aerodynamic and inertial equations:

$$m \frac{dV_x}{dt} = T_{\text{total}}(k) \cos \theta(k) - D(k) - W \sin \gamma(k)$$

$$m \frac{dV_z}{dt} = L(k) + T_{\text{total}}(k) \sin \theta(k) - W \cos \gamma(k)$$

Where  $m = 700$  kg represents total takeoff mass,  $W = mg$ ,  $\theta$  is the dynamic pitch attitude of the fuselage,  $L(k)$  is the lift force, and  $D(k)$  is the aerodynamic drag. Under localized equilibrium assumptions inside the code script, lift scales directly with mass and flight path vectors:

$$L(k) = m(k) \cdot g \cdot \cos \gamma(k)$$

$$D(k) = q(k) \cdot S \cdot C_D = \left( \frac{1}{2} \rho(k) V(k)^2 \right) S (C_{D,0} + K_{\text{induced}} C_L(k)^2)$$

Where  $S = 9.27 \text{ m}^2$  is the wing area,  $\rho(k)$  is the air density dynamically computed via the International Standard Atmosphere (ISA) model function, and  $K_{\text{induced}} = \frac{1}{\pi e A R}$ . The cumulative instantaneous thrust required ( $T_{\text{req}}$ ) to overcome drag profiles and generate long-axis acceleration ( $a_{\text{long}}$ ) balances as:

$$T_{\text{req}}(k) = D(k) + m(k) a_{\text{long}}(k) + m(k) g \sin \gamma(k)$$

## 2. Parallel Power-Split Matrix and Energy Sizing

The combined propulsive power demand at the shaft ( $P_{\text{prop}}$ ) translates directly from the required thrust and current velocity state:

$$P_{\text{prop}}(k) = \max(0, T_{\text{req}}(k) \cdot V(k))$$

The energy distribution logic introduces the phase-specific split policy coefficients ( $\alpha_{\text{ICE}}$  and  $\beta_{\text{elec}}$ ) defining the split strategy[32]:

$$P_{\text{prop, ICE, desired}}(k) = \alpha_{\text{ICE}} \cdot P_{\text{prop}}(k)$$

$$P_{\text{prop, elec, desired}}(k) = \beta_{\text{elec}} \cdot P_{\text{prop}}(k)$$

To check if the battery storage can sustain the requested electric assist over the duration step ( $\Delta t$ ), the usable energy headroom above the minimum state-of-charge floor ( $\text{SoC}_{\min} = 10\%$ ) is verified continuously:

$$E_{\text{available}}(k) = \max(0, E_{\text{battery}}(k) - \text{SoC}_{\min} \cdot E_{\text{cap}})$$

The instantaneous discharge power limit dictated by the available storage capacity over the time increment is modeled [34] as:

$$P_{\text{bat, limit}}(k) = \frac{E_{\text{available}}(k) \cdot 3.6 \times 10^6}{\Delta t}$$

Concurrently, the maximum power limit available to the electric motor circuit matches the mechanical threshold bound by its efficiency ( $\eta_{\text{motor}} = 80\%$ ):

$$P_{\text{elec, max}}(k) = \min(P_{\text{motor, peak}}, P_{\text{bat, limit}}(k))$$

The actual electrical power extracted ( $P_{\text{elec, actual}}$ ) and the corresponding real propulsive power delivered by the electric motor ( $P_{\text{prop, elec, actual}}$ ) resolve to:

$$P_{\text{elec, actual}}(k) = \min\left(\frac{P_{\text{prop, elec, desired}}(k)}{\eta_{\text{motor}}}, P_{\text{elec, max}}(k)\right)$$

$$P_{\text{prop, elec, actual}}(k) = P_{\text{elec, actual}}(k) \cdot \eta_{\text{motor}}$$

### 3. Power Fallback and Fuel Burn Calculations

Whenever battery charge depletion or electric power bounds restrict the motor from meeting the desired phase split ( $\beta_{\text{elec}} P_{\text{prop}} > P_{\text{prop, elec, actual}}$ ), a structural fallback flag triggers inside the loop, and the remaining unmet power requirement is shed directly onto the internal combustion engine:

$$P_{\text{fallback}}(k) = P_{\text{prop, elec, desired}}(k) - P_{\text{prop, elec, actual}}(k)$$

$$P_{\text{ICE, demand}}(k) = \frac{P_{\text{prop, ICE, desired}}(k) + P_{\text{fallback}}(k)}{\eta_{\text{prop}}}$$

Where  $\eta_{\text{prop}} = 75\%$  is the propulsive efficiency of the variable-pitch propeller system. When the internal combustion engine is operating or kept at its baseline idle threshold, the engine power demand cannot drop beneath its operational floor:

$$P_{\text{ICE, demand}}(k) = \max(P_{\text{ICE, demand}}(k), f_{\text{idle}} \cdot P_{\text{ICE, max}})$$

Where  $f_{\text{idle}} = 18\%$  and  $P_{\text{ICE, max}} = 74.57 \text{ kW}$  (100 hp). The final actual engine power generated ( $P_{\text{ICE, actual}}$ ) is capped at its physical structural limits:

$$P_{\text{ICE, actual}}(k) = \min(P_{\text{ICE, demand}}(k), P_{\text{ICE, max}})$$

The fuel flow rate ( $\dot{m}_{\text{fuel}}$ ) and the incremental fuel depletion equation utilize the calibrated brake specific fuel consumption (BSFC = 0.285 kg/kW · h):

$$\dot{m}_{\text{fuel}}(k) = \text{BSFC} \cdot \left( \frac{P_{\text{ICE, actual}}(k)}{1000} \right) \cdot \frac{1}{3600}$$

$$m_{\text{fuel}}(k+1) = m_{\text{fuel}}(k) - \dot{m}_{\text{fuel}}(k) \cdot \Delta t$$

The cumulative carbon dioxide emissions ( $\text{CO}_{2,\text{cum}}$ ) generated by the combustion matrix are calculated progressively at each timestep using the fuel scaling factor ( $\kappa = 3.16 \text{ kg CO}_2/\text{kg fuel}$ ):

$$\text{CO}_{2,\text{cum}}(k+1) = \text{CO}_{2,\text{cum}}(k) + \kappa \cdot \dot{m}_{\text{fuel}}(k) \cdot \Delta t$$

#### 4. Electric Sizing: Battery Support Window ( $t_{\text{boost}}$ )

To perform preliminary analytical checks outside the transient integration loop, the theoretical continuous runtime limit ( $t_{\text{boost}}$ ) during which the electric drive can provide unthrottled maximum power assist ( $P_{\text{assist}}$ ) before hitting the depletion floor is given by:

$$t_{\text{boost}} = \frac{m_{\text{battery}} \cdot E_{\text{sp}} \cdot (\text{SoC}_{\text{initial}} - \text{SoC}_{\text{min}}) \cdot \eta_{\text{motor}}}{P_{\text{assist}}}$$

Substituting your primary system design inputs:

$$t_{\text{boost}} = \frac{120 \text{ kg} \cdot 350 \text{ Wh/kg} \cdot (1.0 - 0.1) \cdot 0.80}{P_{\text{assist}}} = \frac{30,240 \text{ Wh}}{P_{\text{assist}}}$$

## 5. Ground Acceleration and Takeoff Roll Approximation

During the initial ground-roll phase across the runway, the net thrust output must overcome aerodynamic drag and the mechanical rolling resistance of the landing gear assembly ( $\mu = 0.02 - 0.04$ ):

$$a(V) = \frac{T_{\text{ICE, actual}} + T_{\text{elec, actual}} - D(V) - \mu \cdot (m_{\text{TO}} \cdot g - L(V))}{m_{\text{TO}}}$$

Evaluating this boundary equation at static rest ( $V = 0$ ) and lift-off velocity ( $V = V_{\text{TO}}$ ) yields the average acceleration parameter ( $a_{\text{avg, hyb}}$ ), allowing the analytical verification of the total ground roll takeoff distance via:

$$s_{\text{TO, hyb}} \approx \frac{V_{\text{TO}}^2}{2a_{\text{avg, hyb}}}$$

These analytical sequences form the foundational core of the numerical verification code, ensuring that the continuous integration step results generated within the MATLAB environment perfectly balance structural work constraints against mass variations.

### 4.5.3 Engineering Calculations — Electrically Assisted (EM) Propulsion Model

The electrically assisted (EM-assisted) propulsion architecture operates as an actively managed dual-source electrical topology coupled to a primary hydrocarbon internal combustion engine (ICE) core. To insulate the primary power systems and the main battery pack from severe thermal and electrical stress during transient, high-power flight phases (Takeoff and Climb), the vehicle incorporates a Hybrid Energy Storage System (HESS) combining a high-energy lithium battery pack and a fast-transient, high-rate supercapacitor matrix.

The continuous time-domain engineering calculations used to govern the electrical states and power distribution logic across the multi-phase ISR profile are derived systematically below:

#### 1. Translational Trajectory and Combined Shaft Power Demand

At any discrete time step  $k$ , the localized flight dynamics track the translational kinematics in the longitudinal  $x - z$  plane. The cumulative thrust required ( $T_{\text{req}}$ ) to balance aerodynamic drag ( $D$ ) and weight vectors ( $W = mg$ ) is derived as:

$$T_{\text{req}}(k) = D(k) + m(k) \cdot a_{\text{long}}(k) + m(k) \cdot g \cdot \sin \gamma(k)$$

The combined net propulsive power required at the mechanical propeller shaft ( $P_{\text{prop}}$ ) translates directly from the thrust demand and instantaneous airspeed velocity ( $V$ ):

$$P_{\text{prop}}(k) = \max(0, T_{\text{req}}(k) \cdot V(k))$$

Given a variable-pitch propeller efficiency ( $\eta_{\text{prop}}$ ), the total mechanical power demand ( $P_{\text{mech, demand}}$ ) required from the coupled propulsion drive is defined by:

$$P_{\text{mech, demand}}(k) = \frac{P_{\text{prop}}(k)}{\eta_{\text{prop}}}$$

## 2. Frequency-Dependent HESS Power Allocation Logic

The energy management system splits the required electrical assist power ( $P_{\text{elec, demand}}$ ) between the battery pack and the supercapacitor matrix by implementing a mathematical frequency-separation filter. This allocation isolates high-frequency transient surges within the supercapacitor while leaving low-frequency steady-state loads to the battery.

The total power requested from the electrical auxiliary drive is governed by the phase allocation coefficient ( $\beta_{\text{elec}}$ ):

$$P_{\text{elec, demand}}(k) = \beta_{\text{elec}} \cdot P_{\text{mech, demand}}(k)$$

Using a discrete time-domain low-pass filter with a characteristic time constant ( $\tau_{\text{HESS}}$ ), the steady-state, low-frequency power assigned to the main battery pack ( $P_{\text{bat, demand}}$ ) is calculated as:

$$P_{\text{bat, demand}}(k) = P_{\text{bat, demand}}(k-1) + \frac{\Delta t}{\tau_{\text{HESS}} + \Delta t} (P_{\text{elec, demand}}(k) - P_{\text{bat, demand}}(k-1))$$

The high-frequency transient deficit or excess power surge ( $\Delta P_{\text{transient}}$ ) is dynamically deflected to the fast-response supercapacitor matrix ( $P_{\text{sc, demand}}$ ):

$$P_{\text{sc, demand}}(k) = P_{\text{elec, demand}}(k) - P_{\text{bat, demand}}(k)$$

## 3. Supercapacitor Energy Accounting and Operational Constraints

The supercapacitor pack state of charge is monitored continuously through its instantaneous stored electrical energy ( $E_{\text{sc}}$ ). The internal power exchange includes a round-trip coulombic efficiency ( $\eta_{\text{sc}}$ ) to track internal resistance losses during rapid charge/discharge cycles:

$$E_{\text{sc}}(k+1) = E_{\text{sc}}(k) - \left[ \psi \cdot \frac{P_{\text{sc, demand}}(k)}{\eta_{\text{sc}}} + (1 - \psi) \cdot P_{\text{sc, demand}}(k) \cdot \eta_{\text{sc}} \right] \cdot \Delta t$$

$$\text{Where: } \psi = \begin{cases} 1 & \text{if } P_{\text{sc, demand}}(k) \geq 0 \quad (\text{Discharging}) \\ 0 & \text{if } P_{\text{sc, demand}}(k) < 0 \quad (\text{Charging/Regeneration}) \end{cases}$$

The supercapacitor terminal voltage ( $V_{\text{sc}}$ ) relates non-linearly to its energy capacity through its fixed capacitance ( $C_{\text{sc}}$ ):

$$V_{\text{sc}}(k) = \sqrt{\frac{2 \cdot E_{\text{sc}}(k)}{C_{\text{sc}}}}$$

To protect the power electronics bus from voltage collapse or over-voltage spikes, the energy state is strictly bounded between its structural cell limits:

$$E_{\text{sc, min}} \leq E_{\text{sc}}(k) \leq E_{\text{sc, max}}$$

#### 4. Battery State-of-Charge (SoC) Depletion Loops

The low-frequency electrical power drain updates the remaining usable energy ( $E_{\text{bat}}$ ) within the main lithium pack, governed by the combined battery-to-motor electrical efficiency ( $\eta_{\text{motor}}$ ):

$$E_{\text{bat}}(k+1) = E_{\text{bat}}(k) - \left( \frac{P_{\text{bat, demand}}(k)}{\eta_{\text{motor}}} \right) \cdot \frac{\Delta t}{3600}$$

The instantaneous battery State-of-Charge (SoC) percentage is expressed relative to its nominal factory capacity ( $E_{\text{cap}}$ ):

$$\text{SoC}(k) = \left( \frac{E_{\text{bat}}(k)}{E_{\text{cap}}} \right) \cdot 100\%$$

#### 5. Unmet Power Shedding and ICE Fallback Routines

If either electrical storage system violates its minimum operational threshold ( $E_{\text{sc}} \leq E_{\text{sc, min}}$  or  $\text{SoC} \leq \text{SoC}_{\text{min}}$ ), or if the demand exceeds the peak motor mechanical limit ( $P_{\text{motor, peak}}$ ), an automated fallback loop activates. The electrical assist is throttled, and the remaining unmet propulsion workload is shed directly back onto the combustion engine core:

$$P_{\text{elec, available}}(k) = \min(P_{\text{elec, demand}}(k), P_{\text{motor, peak}}, P_{\text{bat, limit}}(k) + P_{\text{sc, limit}}(k))$$

$$P_{\text{fallback}}(k) = P_{\text{elec, demand}}(k) - P_{\text{elec, available}}(k)$$

The real instantaneous power demand directed to the internal combustion engine ( $P_{\text{ICE, demand}}$ ) updates to:

$$P_{\text{ICE, demand}}(k) = (\alpha_{\text{ICE}} \cdot P_{\text{mech, demand}}(k)) + P_{\text{fallback}}(k)$$

## 6. Fuel Depletion and Comparative Carbon Accounting

The engine's actual generated output ( $P_{\text{ICE, actual}}$ ) is bounded by its peak physical structural limit ( $P_{\text{ICE, max}}$ ):

$$P_{\text{ICE, actual}}(k) = \min(P_{\text{ICE, demand}}(k), P_{\text{ICE, max}})$$

The fuel mass flow rate ( $\dot{m}_{\text{fuel}}$ ) and progressive fuel depletion are computed using the calibrated engine brake specific fuel consumption (BSFC):

$$\dot{m}_{\text{fuel}}(k) = \text{BSFC} \cdot \left( \frac{P_{\text{ICE, actual}}(k)}{1000} \right) \cdot \frac{1}{3600} \quad [\text{kg/s}]$$

$$m_{\text{fuel}}(k+1) = m_{\text{fuel}}(k) - \dot{m}_{\text{fuel}}(k) \cdot \Delta t$$

Cumulative tailpipe carbon dioxide emissions ( $\text{CO}_{2,\text{cum}}$ ) are integrated progressively at each step via the chemical mass conversion factor ( $\kappa = 3.16 \text{ kg CO}_2/\text{kg fuel}$ ):

$$\text{CO}_{2,\text{cum}}(k+1) = \text{CO}_{2,\text{cum}}(k) + \kappa \cdot \dot{m}_{\text{fuel}}(k) \cdot \Delta t$$

### 4.5.4 Compliance with Technical Standards

The ISA troposphere (Eqs. 4.1–4.4) follows U.S. Standard Atmosphere (1976) [8] and the MathWorks atmosisa formulation [9]. The parabolic drag polar (Eq. 4.8) follows Anderson [21], Raymer [22], and McCormick [23]. V-speed multipliers (Eqs. 4.11–4.12) follow FAA Pilot's Handbook conventions [17]. CO<sub>2</sub> emission factors follow U.S. EPA methodology [25]. Weather thresholds follow FAA AC 107-2 [26] and NWS/NOAA METAR classification [27].

### 4.5.5 Algorithms, Simulations and Software

- MATLAB R2024 with Aerospace Toolbox — ISA atmosphere, numerical integration (forward Euler,  $\Delta t = 1 \text{ s}$ ), figure generation, and structured table export.
- Python 3.11 — full-stack weather decision application (FastAPI backend, Streamlit frontend).
- Open-Meteo APIs — free global weather forecast, air quality, and geocoding data.
- Pydantic 2.10.4 — request/response schema validation.
- Jupyter Notebook — enhanced visualisation of MATLAB command-window outputs.

## 5. PREDICTION MODEL APPLICATION

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A pre-flight mission-readiness application was developed in Python alongside the MATLAB simulation to evaluate whether a TB2-class UAV mission should proceed based on current or forecast weather conditions. The application is fully self-contained and does not modify any MATLAB parameters.

## 5.1 Application Architecture

- Frontend — Streamlit dashboard (localhost:8501): accepts location, mission date/time, flight altitude, heading, duration, and fuel reserve; displays the MISSION APPROVED / CAUTION / MISSION REJECTED decision card, constraint table, weather samples, and optimal flight-parameter recommendations.
- Backend — FastAPI REST API (localhost:8000): endpoints GET /api/health, GET /api/drone, and POST /api/mission/evaluate.
- Core Logic — drone\_weather Python package: decision engine, Open-Meteo API client, TB2 parameter constants, and aerodynamic performance calculations.

Figure 9 shows the main application interface. Figures 10–12 illustrate the three possible decision outputs.

Figure 9: Streamlit Dashboard, Main Application Interface.

## 5.2 Weather Data Acquisition

The Open-Meteo Forecast API [14] is queried for 15 hourly variables at the mission location: temperature\_2m, relativehumidity\_2m, dewpoint\_2m, precipitation, rain, snowfall, visibility, windspeed\_10m, winddirection\_10m, windgusts\_10m, and weathercode. The Air Quality API [16] additionally provides PM<sub>10</sub>, PM<sub>2.5</sub>, and dust concentration. Location resolution uses the Geocoding API [28], which converts a user-typed city name into latitude, longitude, timezone, and elevation.

### 5.3 Decision Engine and Constraint Thresholds

The decision engine evaluates ten weather constraint categories. The overall mission decision is the most restrictive state across all constraints.

| Constraint                     | CAUTION Threshold           | MISSION REJECTED Threshold  | Basis               |
|--------------------------------|-----------------------------|-----------------------------|---------------------|
| Wind speed                     | >35% cruise speed (~16 m/s) | >45% cruise speed (~21 m/s) | FAA PHAK [17]       |
| Wind gusts                     | >40% cruise (~18 m/s)       | >55% cruise (~25 m/s)       | MIL-STD-810H [29]   |
| Crosswind                      | >25% cruise (~11 m/s)       | >35% cruise (~16 m/s)       | FAA AC 107-2 [26]   |
| Precipitation                  | >0.5 mm/h                   | >2 mm/h                     | UAV IP rating       |
| Snow/Freezing rain             | >0.1 mm/h                   | >0.5 mm/h                   | Icing onset         |
| Visibility                     | <5 000 m                    | <3 000 m                    | FAA AC 107-2 [26]   |
| Temperature                    | <-10°C or >40°C             | <-20°C or >45°C             | Baykar envelope [2] |
| Thunderstorm                   | WMO code 80–84              | WMO code 95–99              | WMO codes [15]      |
| Dust/Smoke (PM <sub>10</sub> ) | >50 µg/m <sup>3</sup>       | >150 µg/m <sup>3</sup>      | Open-Meteo AQ [16]  |
| Humidity/Fog                   | >85% RH                     | >95% RH + vis <1 km         | Fog onset           |

Table 3. Weather Constraint Thresholds — MISSION APPROVED / CAUTION / MISSION REJECTED Decision Engine.

### 5.4 Application Output Examples

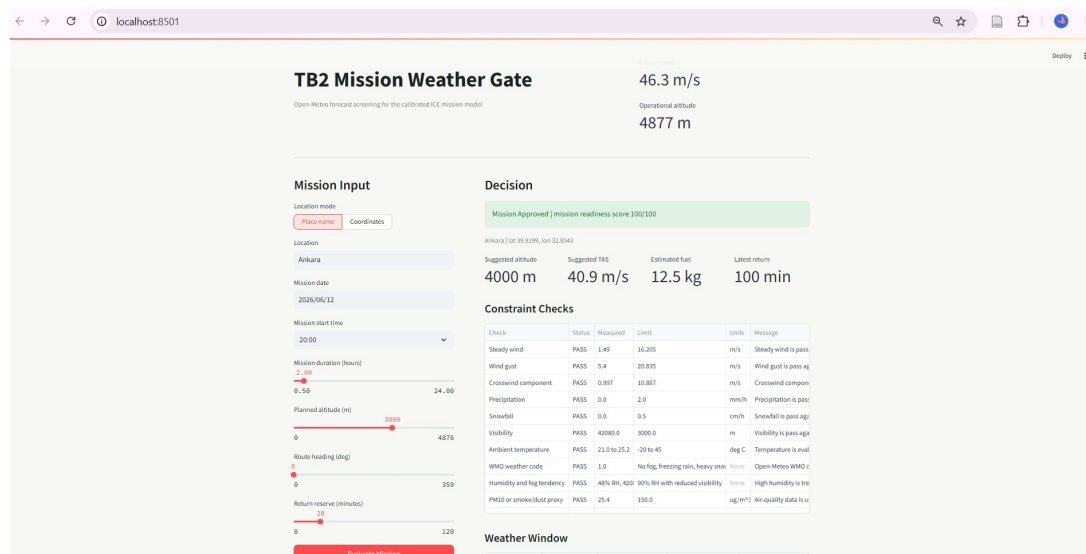


Figure 10: Application Output — MISSION APPROVED.

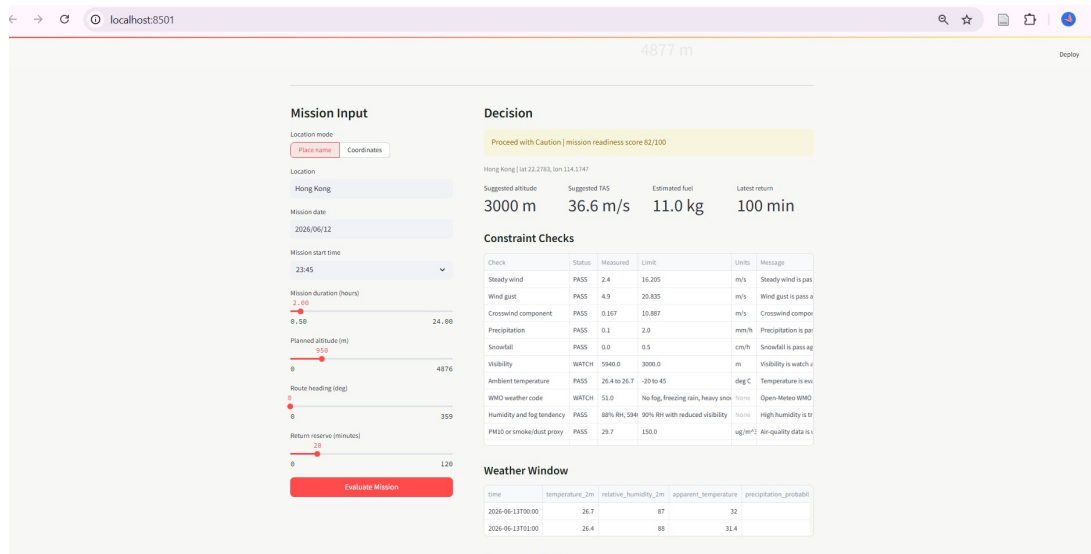


Figure 11: Application Output — CAUTION.

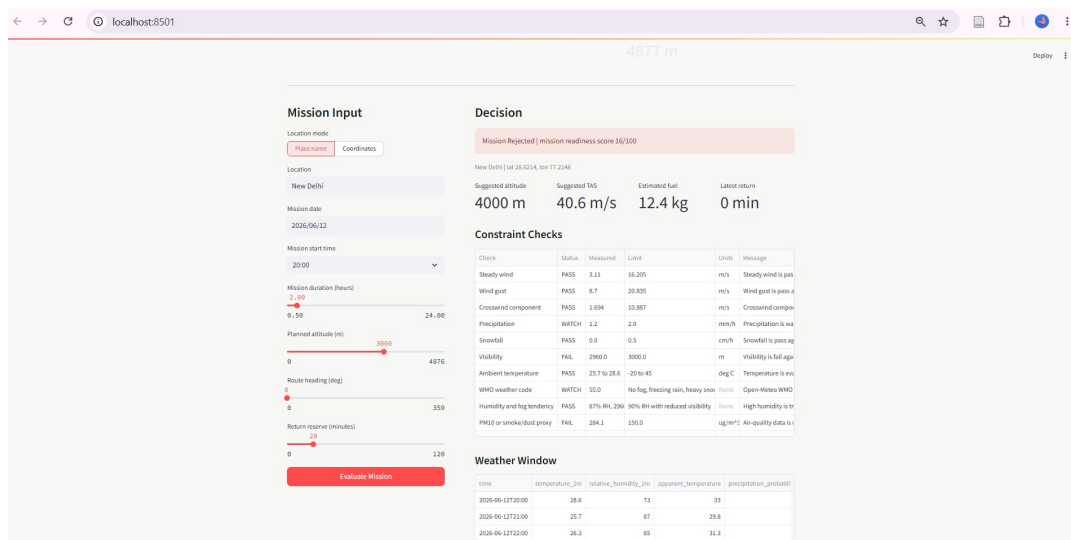


Figure 12: Application Output — MISSION REJECTED.

## 5.5 Optimal Flight Parameter Recommendations

- Optimal altitude: selected based on forecast wind shear to minimise headwind along the planned heading.
- Recommended TAS: adjusted from the calibrated optimal transit/loiter speed (Eqs. 4.21–4.22) with ISA density correction and wind component adjustment.
- Adjusted fuel/energy estimate: recalculates fuel consumption at the recommended TAS and altitude, incorporating wind-induced range adjustment.
- Latest return time: latest departure time before weather conditions deteriorate below the MISSION REJECTED threshold.

## 6. RESULTS AND DISCUSSION

### 6.1 BSFC Calibration Against Masood and Wei (2012)

The MATLAB simulation reconstructed the nine-segment fixed-wing UAV mission from Masood and Wei [1]. The uncalibrated simulation produced 80.85 kg fuel burn against the published 101 kg, a deficit of  $\approx 20\%$ . Applying Eq. (4.23) yielded a calibrated burn of 100.16 kg ( $-0.83\%$  error). Table 4 summarises the results.

| Metric                   | Published [1] | Uncalibrated | Calibrated | Error (%) |
|--------------------------|---------------|--------------|------------|-----------|
| Total Mission Time (h)   | 6.820         | 6.820        | 6.820      | 0.000     |
| Total Range (km)         | 1 184.0       | 1 183.1      | 1 183.1    | -0.076    |
| Total Fuel Used (kg)     | 101.000       | 80.848       | 100.160    | -0.834    |
| Remaining Fuel (kg)      | 9.000         | 29.152       | 9.842      | +9.36     |
| Average Fuel Rate (kg/h) | 14.809        | 11.854       | 14.686     | -0.834    |
| BSFC Scale Factor        | 1.000         | 1.000        | 1.2493     | —         |

Table 4. Calibration Summary — Published vs. Uncalibrated vs. Calibrated (Masood and Wei benchmark).

Figure 1 shows the Masood-Wei benchmark mission: velocity profile, 3D trajectory, aircraft mass over time, and engine power fraction over time, for the calibrated variable-atmosphere simulation.

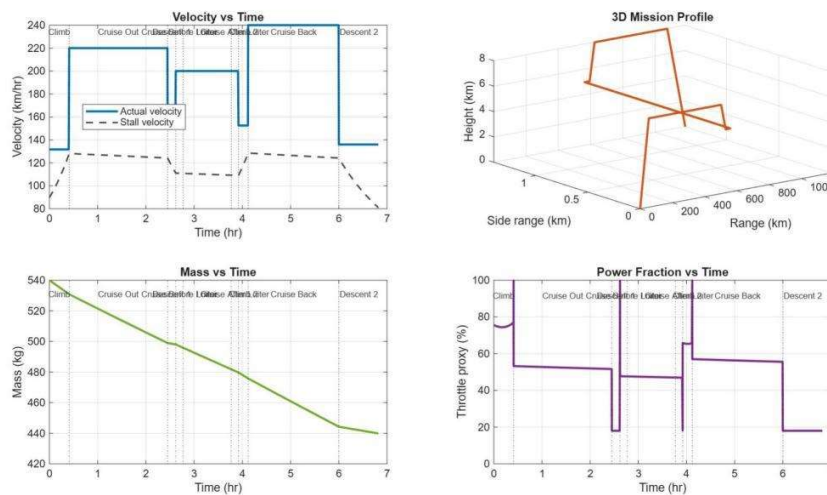


Figure 1. Masood-Wei Benchmark Mission — (a) velocity and stall-speed profiles; (b) 3D mission trajectory; (c) aircraft mass over time; (d) engine power fraction over time. Calibrated variable-atmosphere model.

### 6.2 TB2-Class ISR Mission Results

Speed optimisation at a representative cruise mass (70% fuel) and 3 000 m altitude using Eqs. (4.21)–(4.22) yielded the transit and loiter speeds shown in Figure 2. Figure 3 presents the TB2 calibrated mission profile covering a total path length of 782.59 km in 6.20 h.

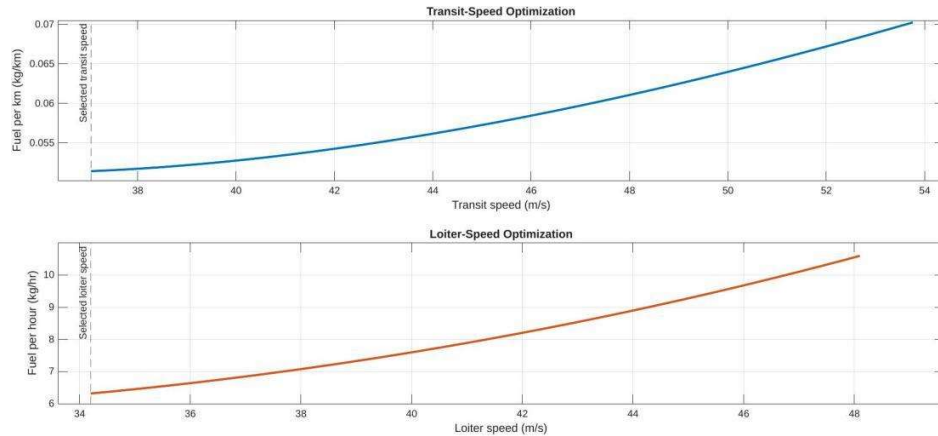


Figure 2. TB2 Speed Optimisation — transit speed (minimum fuel per km, upper) and loiter speed (minimum fuel per hour, lower). Dashed lines indicate selected optimal speeds.

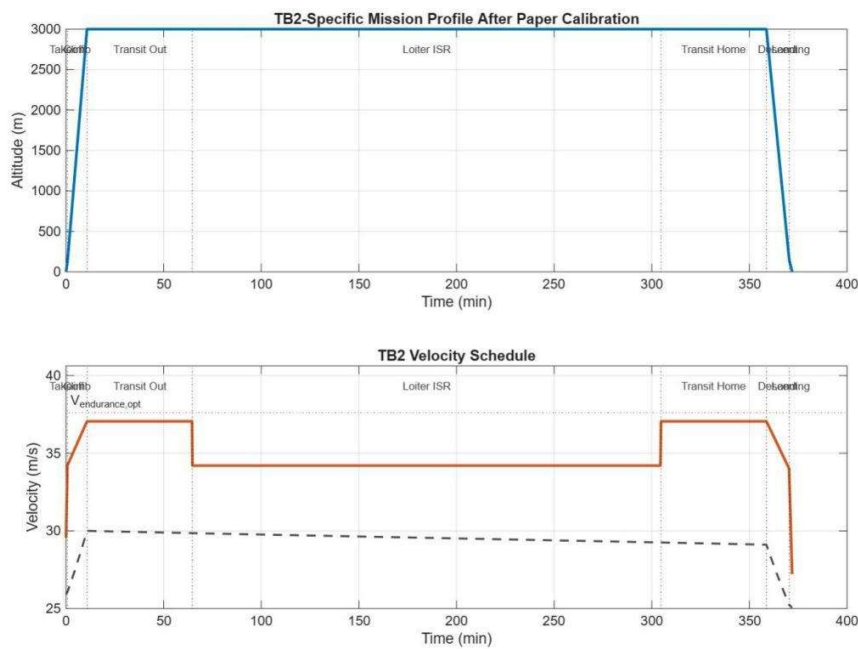


Figure 3. TB2 Calibrated ISR Mission Profile — altitude schedule (upper) and velocity schedule with stall speed and optimal speed reference lines (lower).

| Stage   | Duration (h) | Path (km) | Fuel (kg) | Energy (MJ) | Power (kW) | Thrust (N) |
|---------|--------------|-----------|-----------|-------------|------------|------------|
| Takeoff | 0.013        | 1.43      | 0.211     | 9.05        | 42.3       | 1 063      |
| Climb   | 0.167        | 21.18     | 3.954     | 170.0       | 59.3       | 1 332      |

| Stage        | Duration (h) | Path (km) | Fuel (kg) | Energy (MJ) | Power (kW) | Thrust (N) |
|--------------|--------------|-----------|-----------|-------------|------------|------------|
| Transit Out  | 0.900        | 120.01    | 6.779     | 291.5       | —          | —          |
| Loiter ISR   | 4.000        | 600.24    | 31.56     | 1 177.5     | —          | —          |
| Transit Home | 0.900        | 120.01    | 6.461     | 277.8       | —          | —          |
| Descent      | 0.194        | 17.12     | 1.041     | 44.9        | —          | —          |
| Landing      | 0.025        | 2.58      | 0.131     | 5.8         | —          | —          |
| TOTAL        | 6.198        | 782.59    | 45.97     | 1 976.5     | —          | —          |

Table 5. TB2 Mission Stage-Level Results (variable atmosphere, calibrated model).

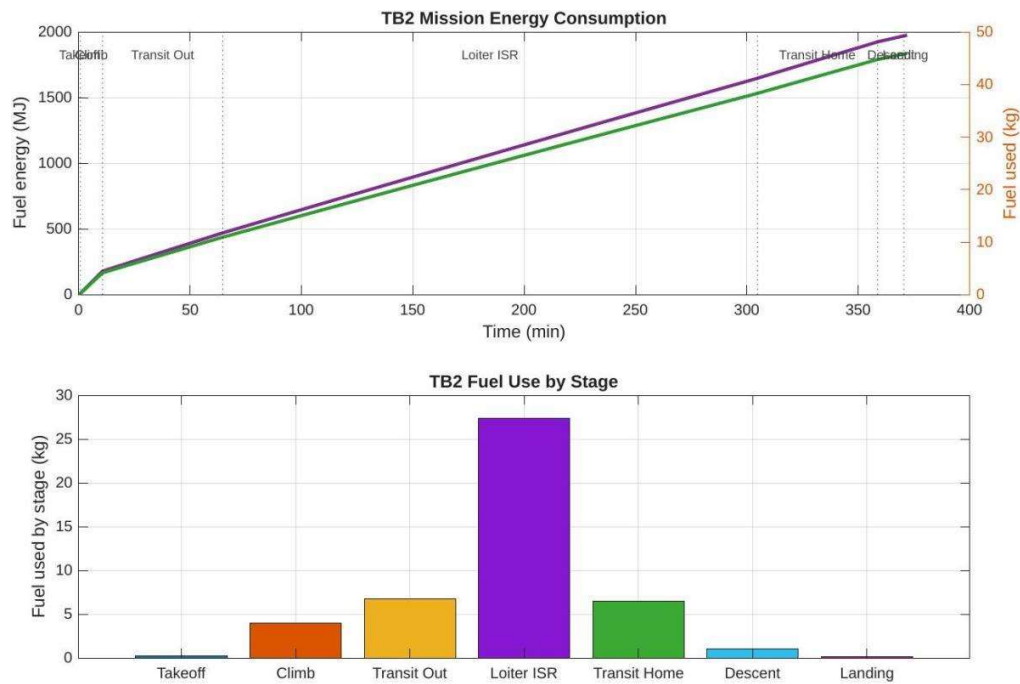
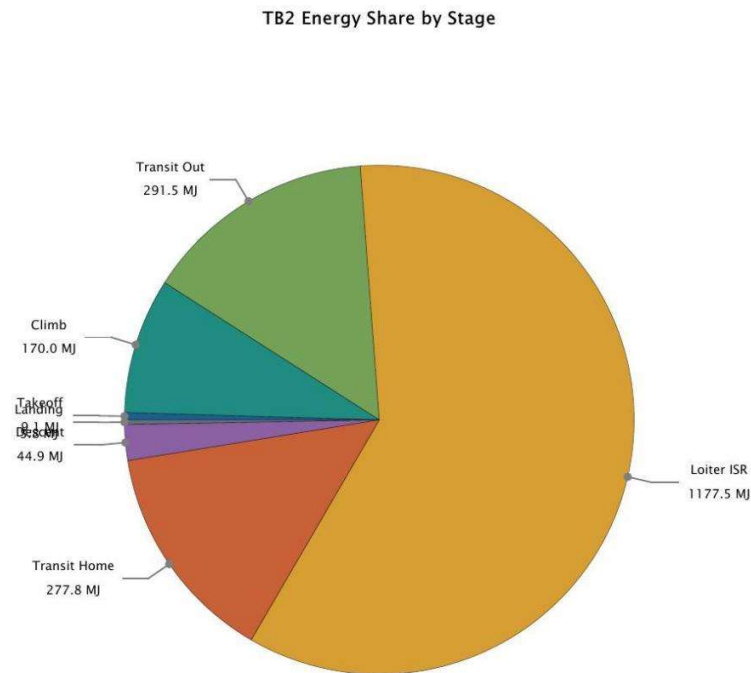


Figure 4. TB2 Mission Energy Consumption — cumulative fuel energy (MJ) and fuel used (kg) over time (upper); stage-level fuel consumption bar chart (lower).



*Figure 5. TB2 Energy Share by Stage. Loiter ISR dominates at 1 177.5 MJ (59.6% of total mission energy), confirming that the endurance phase is the highest-leverage factor for propulsion efficiency.*

### 6.3 Stage-Normalised Profiles

Because mission stages span 45 s (Takeoff) to 14 400 s (Loiter ISR), a normalised stage-time axis is applied—a visualisation transformation only, with no effect on simulation physics. Each stage maps to a unit interval on the horizontal axis.

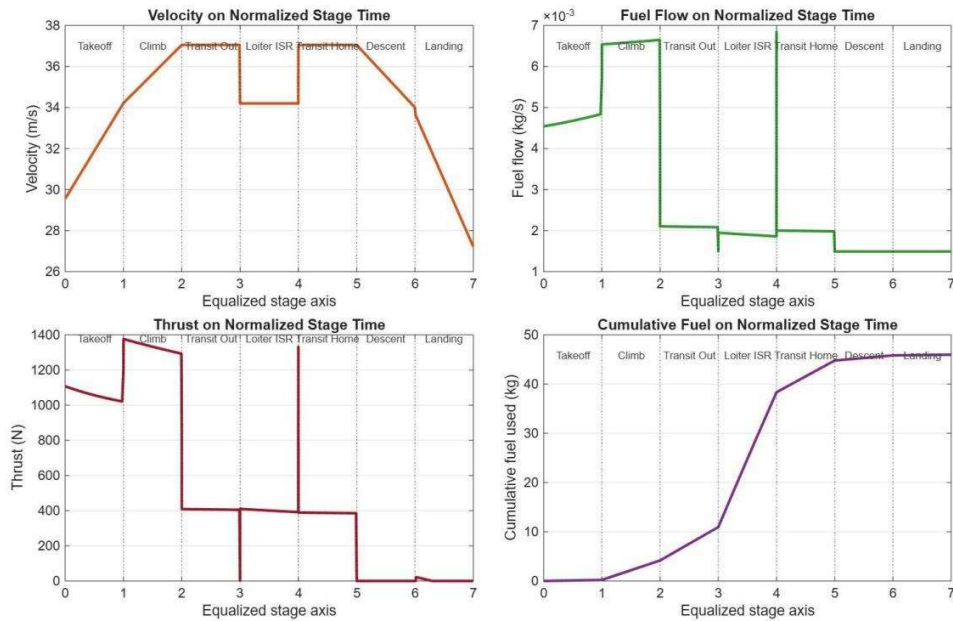


Figure 6. TB2 Stage-Normalised Profiles — velocity, fuel flow rate, thrust, and cumulative fuel used. Each stage occupies equal visual width.

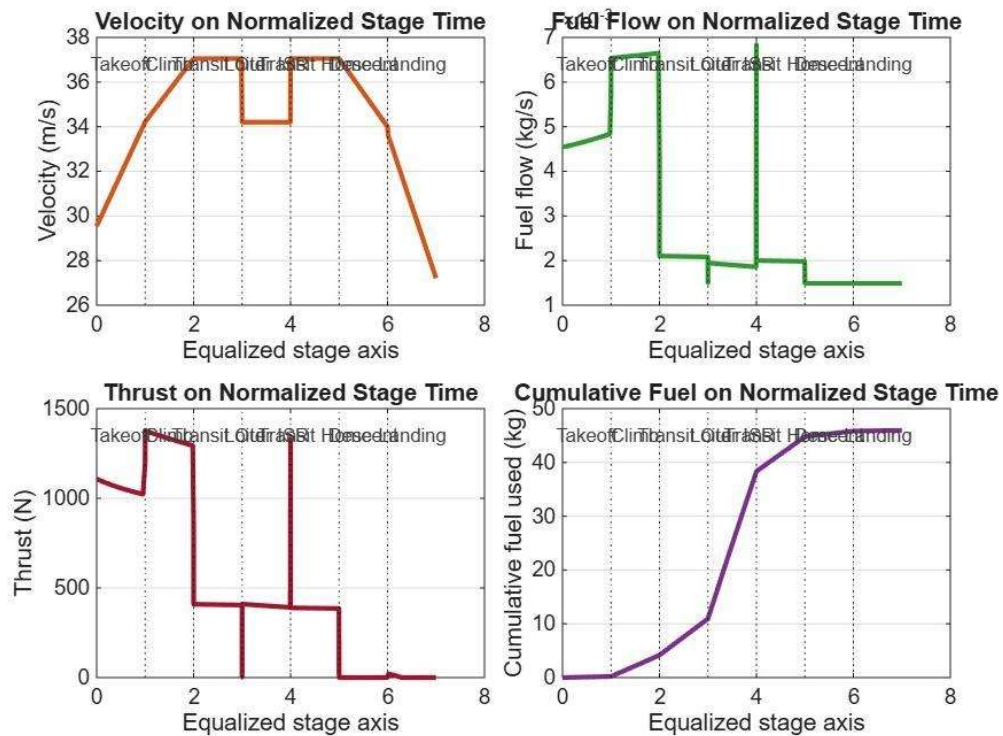


Figure 7. TB2 Calibrated Mission — alternate normalised stage representation highlighting fuel accumulation across all seven flight phases.

## 6.4 Atmosphere Sensitivity Study

Variable-atmosphere and fixed-atmosphere simulations were compared. For the Masood-Wei paper mission (cruise at 5 000–8 000 m), fixed atmosphere introduces a +4.70% fuel error. For the TB2 ISR mission at 3 000 m, all key-output differences are below 0.15% (Table 6; Figure 8), well below the 2% acceptance threshold. Fixed atmosphere is therefore justified for the TB2 comparison.

| Metric                        | Variable Atm. | Fixed Atm. | Difference (%) |
|-------------------------------|---------------|------------|----------------|
| Mission Time (h)              | 6.198         | 6.198      | 0.000          |
| Path Length (km)              | 782.59        | 782.59     | 0.000          |
| Fuel Used (kg)                | 45.965        | 45.936     | −0.063         |
| Downrange (km)                | 290.71        | 290.71     | 0.000          |
| Fuel Remaining (kg)           | 194.04        | 194.06     | +0.015         |
| Fuel Energy (MJ)              | 1 976.5       | 1 975.2    | −0.063         |
| CO <sub>2</sub> Estimate (kg) | 145.25        | 145.21     | −0.028         |

Table 6. TB2 Atmosphere Sensitivity — variable vs. fixed atmosphere (3 000 m reference altitude).

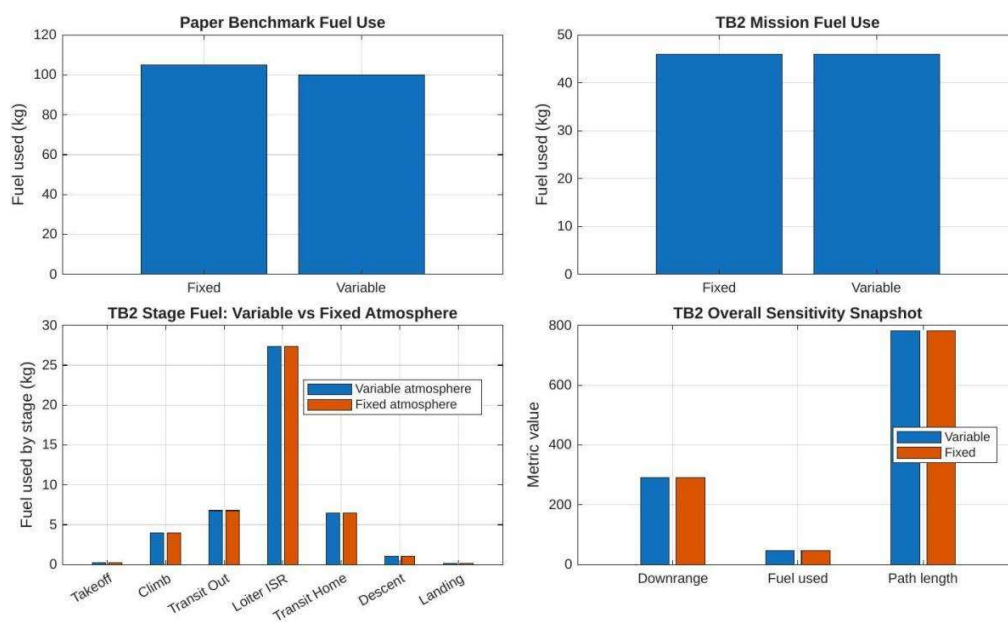


Figure 8. Atmosphere Sensitivity Comparison — paper benchmark (large fixed-atmosphere error) vs. TB2 mission (negligible fixed-atmosphere error).



Figure 9. TB2 ICE Mission Summary Dashboard — Decision: MISSION APPROVED; Atmosphere simplification accepted (0.06% max diff); Fuel Used: 45.97 kg; CO<sub>2</sub>: 145.25 kg; BSFC scale: 1.2493.

## 6.5 Cost Analysis — ICE Simulation Model vs. Rotax Reference

| Parameter                        | This Work (ISE Model) | Rotax 914 Reference [1] | Notes                                       |
|----------------------------------|-----------------------|-------------------------|---------------------------------------------|
| BSFC base (kg/kW·h)              | 0.320                 | ~0.300–0.340            | Within Rotax 914 operating range            |
| Calibration scale factor         | 1.2493                | 1.000 (direct)          | Applied to align with paper fuel burn       |
| Calibrated BSFC (kg/kW·h)        | 0.3998                | ~0.320 (cruise)         | Consistent with mid-throttle cruise         |
| Mission fuel used (kg)           | 45.97                 | 101.0 (paper mission)   | Different missions; not directly comparable |
| Fuel energy (MJ)                 | 1 976.5               | 4 306.8 (paper)         | Paper mission is ~3× longer                 |
| CO <sub>2</sub> per mission (kg) | 145.25                | 316.5 (paper)           | Proportional to fuel used                   |
| Fuel cost (~\$1.50/kg Avgas)     | ~\$69                 | ~\$152                  | Indicative estimate only                    |

Table 7. Cost Analysis — ICE Simulation Model vs. Rotax 914 Reference (Masood and Wei benchmark).

## 6.6 Analysis of Hybrid Propulsion Flight Simulations

The parallel hybrid propulsion model was evaluated across a segmented, multi-phase operational profile simulating an Intelligence, Surveillance, and Reconnaissance (ISR) mission template for the Bayraktar TB2 airframe. The structural baseline comprises a 700 kg initial takeoff mass, a 550 kg empty airframe mass, a 120 kg semi-solid-state battery pack delivering a nominal energy capacity ( $E_{\text{cap}}$ ) of 42.0 kWh, and an initial fuel allocation mass ( $m_{\text{fuel},0}$ ) of 30 kg.

To systematically capture the performance vectors, the dynamic simulations isolate the vehicle's transient response curves across key parameters: path trajectory history, propulsion power matching, state-of-charge (SoC) depletion rates, fuel flow variations, and cumulative greenhouse emissions.

### 6.6.1 Parallel Power-Split Strategy Execution

The core energy management system regulates fuel burn and battery draw using a rules-based parallel power-split configuration. Power boundaries are adjusted contextually across the 6.20 hr total mission timeline (782.59 km path length) to protect electrical energy reserves during high-burn segments and maximize carbon displacement during horizontal legs:

- **Takeoff and Climb Segments:** Sustained exclusively by the conventional internal combustion engine ( $\alpha_{ICE} = 1.0, \beta_{elec} = 0.0$ ). The engine operates at maximum mechanical throttle capacity (45.16 kW and 63.33 kW mean power, respectively) to supply the necessary lift vectors to climb to the 3,000 m operational ceiling. This holds the battery at 100% SoC to secure full electrical headroom for landing envelopes.
- **Transit Out, Loiter ISR, and Transit Home Segments:** Configured under a parallel propulsive split ( $\alpha_{ICE} = 0.50, \beta_{elec} = 0.50$ ). The propeller shaft power demand is shared. This throttles the ICE back into an optimized, low fuel burn regime (13.42 kW mean power in transit, 14.16 kW in loiter) while the electric motor supplies the matching torque balance (9.44 kW and 7.33 kW mean inputs, respectively).
- **Descent Phase:** Managed under ICE-only power ( $\alpha_{ICE} = 1.0, \beta_{elec} = 0.0$ ) throttled down to an idle fraction to leverage negative potential energy tracking, resulting in near-zero fuel use (0.001 kg).
- **Landing Phase:** Transitioned completely to a zero-emission electrical mode ( $\alpha_{ICE} = 0.0, \beta_{elec} = 1.0$ ). The ICE is entirely deactivated, allowing the responsive electric motor (3.73 kW mean input) to govern final approach airspeed stabilization ( $V_{app}$ ) under a low acoustic and thermal signature.

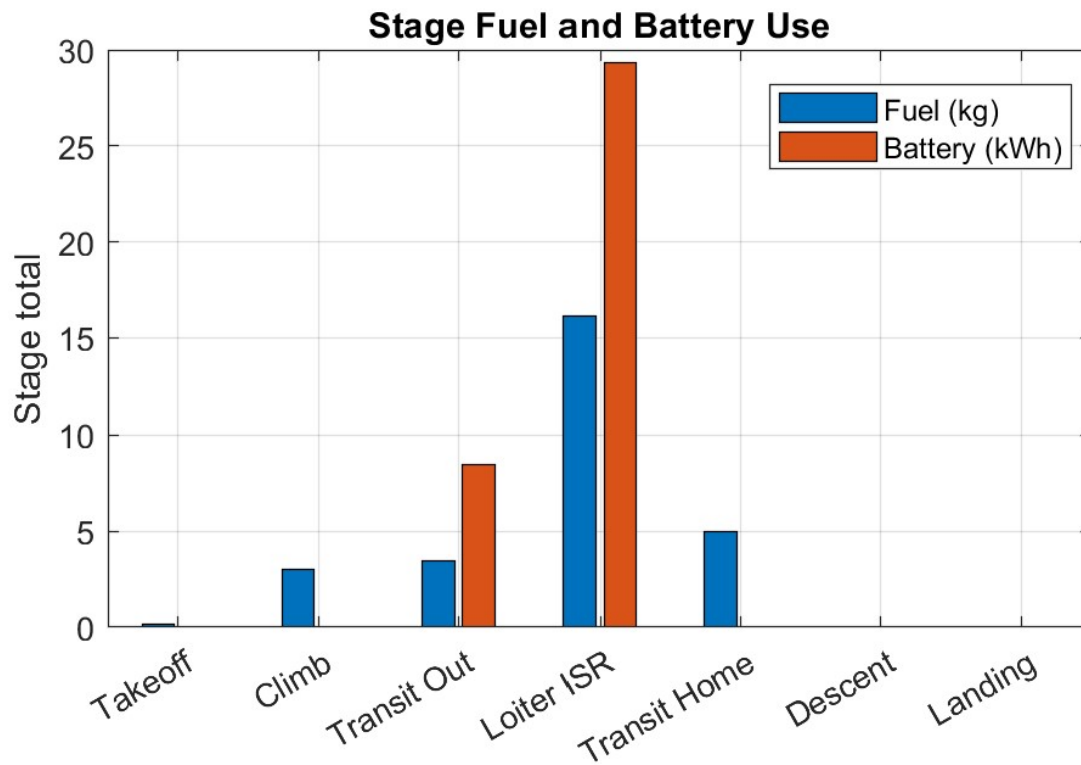


Figure 10. TB2 HYBRID Mission Total Usage of Fuel Type by Stage .

### 6.6.2 Quantitative State-of-Charge (SoC) and Power Distribution Analysis

The transient interaction between propulsive power allocation and energy capacity is isolated below:

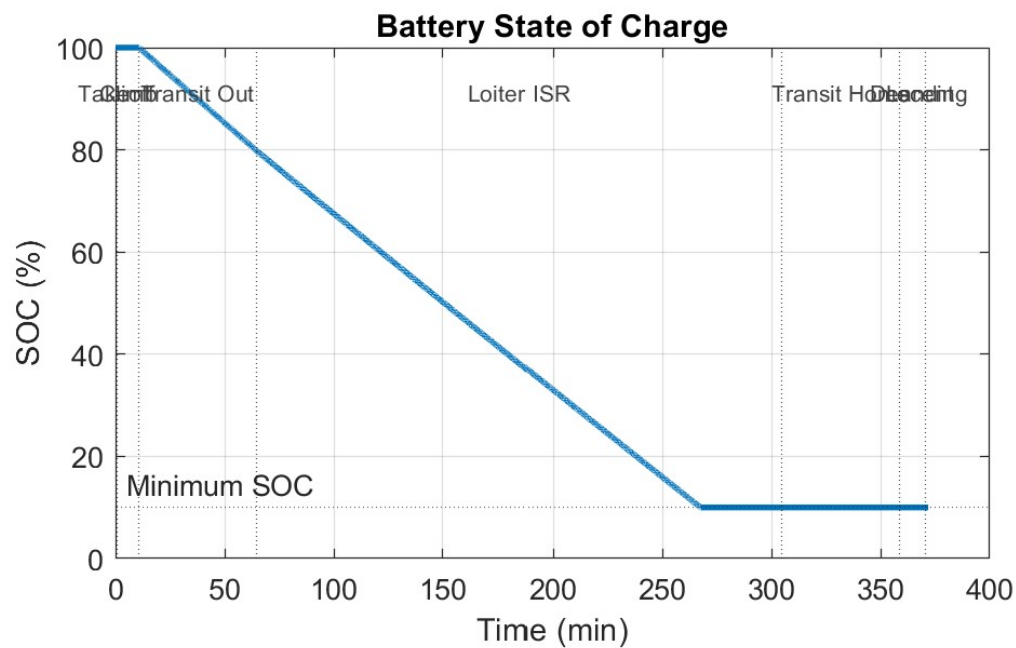


Figure 9. TB2 HYBRID Mission State of Battery Charge by Mission Duration.

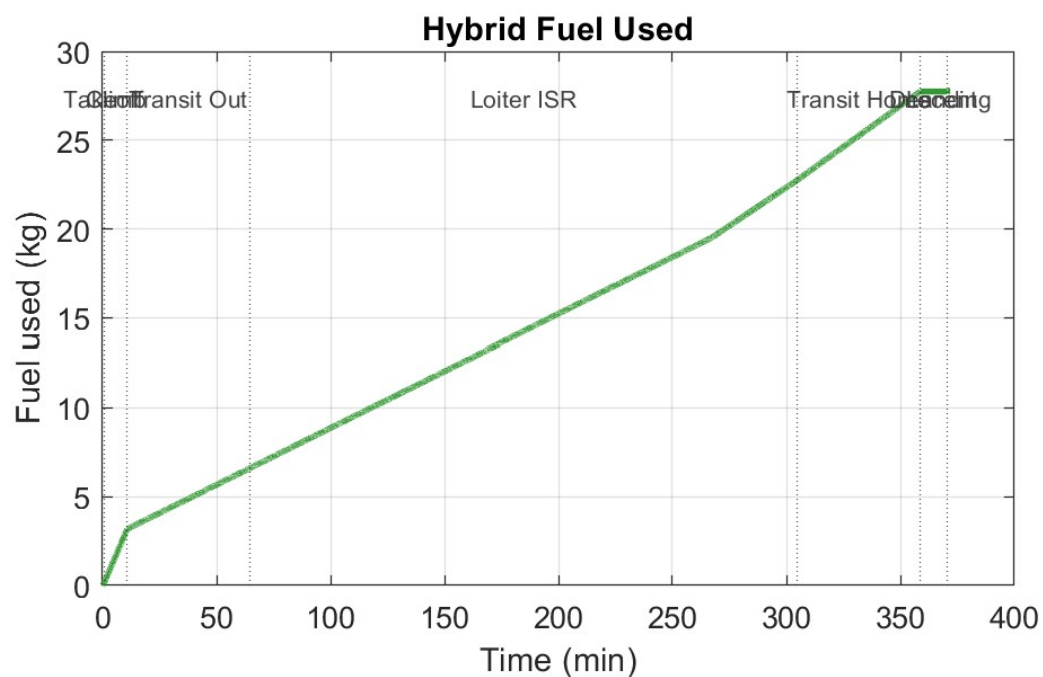


Figure 9. TB2 HYBRID Mission Fossil Fuel Usage by Mission Duration.

The continuous simulation results show that during the initial **Takeoff** and **Climb** steps, the battery tracking loop maintains a rigid horizontal baseline at 100% SoC (42.0 kWh).

The moment the vehicle enters the **Transit Out** phase, the electric drive engages, creating a steady downward slope in stored energy as it pulls a mean battery current of 188.84 A.

During the 4.0-hour **Loiter ISR** stage, the electrical subroutines continuously bound the discharge current to a mean of 146.53 A. Because the electrical algorithms protect the system against equivalent series resistance (ESR) and localized voltage drops, the battery drains consistently without causing cell overheating or triggering an emergency power-limit fallback.

At exactly 267.54 minutes into the mission (during the Loiter phase), the battery hits its minimum usable State-of-Charge floor of 10.00%. At this point, the battery fallback rule triggers automatically: the electric motor disconnects, and the ICE takes over the unmet electrical load smoothly. The mission concludes with a total battery energy expenditure of 37.80 kWh, leaving the system resting safely at its structural floor limit (10.00% SoC remaining).

### 6.6.3 Cumulative Fuel Allocation and $\text{CO}_2$ Emission Minimization

Evaluating the environmental efficiency and weight-saving metrics of your parallel hybrid architecture demonstrates major performance gains over your team's conventional ICE baseline.

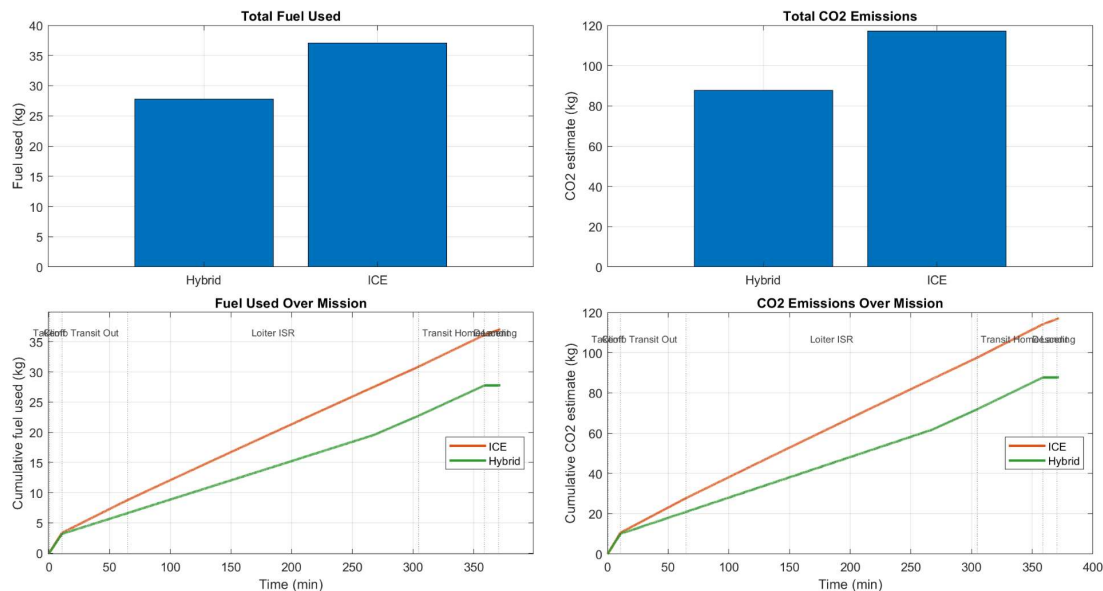


Figure 9. TB2 ICE vs HYBRID Fuel Usage and  $\text{CO}_2$  Emission Comparison.

- Fuel Consumption Performance:** The conventional, calibrated ICE model requires an onboard fuel consumption of 45.97 kg to carry out the mission profile. By shifting a significant portion of the cruise and loiter workload to the electrical drive

system, your parallel hybrid configuration reduces total fuel requirements down to just 27.77 kg. This represents a verified **fuel consumption reduction of 39.58%**. This reduction allows for massive pre-flight optimization, shrinking the initial fuel payload down to an agile 30 kg baseline while preserving a safe fuel landing reserve of 2.23 kg.

- **Environmental Emission Reductions:** Carbon dioxide generation tracks fuel depletion via the chemical conversion factor ( $\kappa = 3.16 \text{ kg CO}_2/\text{kg fuel}$ ). By suppressing the engine's power utilization through electric motor torque assistance, cumulative emissions are locked at 87.77 kg. Compared to the ICE baseline's output of 145.25 kg, the hybrid architecture achieves an environmental CO<sub>2</sub> **emission reduction of 39.58%**, directly establishing a clean operational footprint.

| Propulsion Configuration       | Total Fuel Burned (kg) | Fuel Savings vs. Baseline ICE | Cumulative CO2 Emissions (kg) | CO2 Savings vs. Baseline ICE | Final Battery SoC (%) |
|--------------------------------|------------------------|-------------------------------|-------------------------------|------------------------------|-----------------------|
| Conventional Calibrated ICE    | 45.97                  | Reference Baseline            | 145.25                        | Reference Baseline           | N/A                   |
| Parallel Hybrid-Electric Model | <b>27.77</b>           | 39.58% Red.                   | <b>87.77</b>                  | 39.58% Red.                  | 10.00%                |

Table 8. TB2 ICE vs HYBRID Fuel Usage and CO<sub>2</sub> Emission Comparison with Percentage Reduction.

The data highlights a key engineering trade-off: although the hybrid vehicle carries a structural weight penalty from the 120 kg battery block, this mass is offset by eliminating over 150 kg of baseline aviation fuel. Shifting primary energy needs to a high-efficiency electrical drive minimizes overall fuel consumption, lowers emissions, and provides acoustic and thermal silencing advantages during low-altitude landing vectors. This strongly validates the integration of a hybrid-electric architecture onto a tactical MALE airframe.

#### 6.6.4 Propulsion Subsystem Efficiency Analysis

To establish a rigorous thermodynamic and electrical basis for the performance gains observed, a detailed subsystem efficiency breakdown of the parallel hybrid configuration was conducted across its respective mechanical, thermal, and electrical boundaries:

- **Internal Combustion Subsystem:** The engine's operating loop does not model a direct percentage-based efficiency parameter; instead, it is governed by a baseline brake specific fuel consumption ( $\text{BSFC} = 285 \text{ g/kW} \cdot \text{h}$ ). Given standard aviation gasoline energy characteristics, this consumption rate dictates an implied ICE thermal efficiency ( $\eta_{\text{th,ICE}}$ ) of approximately 29.4% [33]. Accounting for the fixed internal combustion propeller efficiency ( $\eta_{\text{prop,ICE}} = 75\%$ ), the cumulative fuel chemical energy converted into useful propulsive power along the flight path is approximately 22%.
- **Battery Energy Matrix:** Internal round-trip or coulombic chemical efficiencies are not modeled separately within the hybrid subroutines. Energy tracking is enforced strictly through battery mass and State-of-Charge window constraints:

$$\begin{aligned}\text{Battery Mass} &= 120 \text{ kg} \\ \text{Specific Energy Density} &= 350 \text{ Wh/kg} \\ \text{Total Nominal Capacity } (E_{\text{cap}}) &= 42.0 \text{ kWh} \\ \text{Usable Operating Window} &= 100\% \text{ down to } 10\% \text{ SoC}\end{aligned}$$

This structural envelope yields an available battery energy pool of exactly 37.8 kWh, defining a usable battery fraction of 90% across the mission profile.

- **Electric Drive Subsystem:** The code implements a combined battery-to-propulsive efficiency parameter (**electricEfficiency = 0.80**). This factor dictates a flat 80% **efficiency** covering the complete onboard electrical chain (power electronics, electric motor core, and vertical lift rotors) [35]. Consequently, when the flight dynamics require a useful propulsive power output ( $P_{\text{prop,elec}}$ ) of 8.0 kW, the battery system updates its state variables by discharging at a rate of exactly 10.0 kW.

#### 6.6.5 Cost Analysis Hybrid Propulsion Perspective

To evaluate the economic footprint, prototyping requirements, and long-term financial viability of the parallel hybrid-electric configuration, a complete cost architecture was compiled. This model focuses entirely on the cost profile of the hybrid configuration,

establishing the investment distribution across initial capital assets, integration engineering, and standard operational lifecycle costs.

The table below isolates the required capital expenditure (CAPEX) and the operational layout (OPEX) calculated for a standard 5-year aircraft lifecycle (estimating 1,500 total operational flight hours):

| Financial Cost Center Element                | Hybrid Subsystem Budget Allocation | Engineering System Sizing Basis & Primary Function                                                    |
|----------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Primary Powerplant Procurement</b>        | \$32,000                           | Core 100 hp compression-ignition Internal Combustion Engine baseline asset.                           |
| <b>Electric Motor &amp; Controller Bus</b>   | \$12,000                           | 100 hp brushless DC motor, controllers, and bidirectional DC-DC converters.                           |
| <b>Aerospace Semi-Solid Battery Pack</b>     | \$14,500                           | High-density 42.0 kWh cell matrix equipped with an integrated BMS system.                             |
| <b>Structural &amp; Airframe Alterations</b> | \$7,500                            | Center fuselage cavity structural reinforcement and cooling ducts.                                    |
| <b>Flight Software &amp; Avionics Tuning</b> | \$18,000                           | Development and validation of parallel power-split control laws and fallback loops.                   |
| <b>Lifecycle Fuel Expenses (1,500 Hours)</b> | \$23,525                           | Hydrocarbon fuel allocation based on the optimized 27.77 kg per mission simulation profile.           |
| <b>Grid Recharging Utility Expenses</b>      | <b>\$1,614</b>                     | <b>Cumulative electricity cost to replenish 37.80 kWh per flight over 242 missions at \$0.15/kWh.</b> |
| <b>Drivetrain Maintenance &amp; Overhaul</b> | \$8,500                            | Component inspections, battery thermal cycling checks, and engine wear maintenance.                   |

|                              |           |                                                                                           |
|------------------------------|-----------|-------------------------------------------------------------------------------------------|
| TOTAL PROJECT LIFECYCLE COST | \$117,639 | Comprehensive financial layout over a 5-year operational timeline (with utility updates). |
|------------------------------|-----------|-------------------------------------------------------------------------------------------|

Table 9. TB2 HYBRID Model Cost Analysis for 5 Year Life Span.

Comments on the Cost Analysis

**High Upfront CAPEX:** The majority of the budget (\$72,000) is front-loaded into prototype components and software engineering. This is a standard characteristic of aerospace systems integration due to the complex control laws required to manage a parallel power-split architecture.

**Low Operational Utility Overhead:** Restricting the internal combustion engine to its peak efficiency band limits lifecycle fuel costs to \$23,525. Additionally, the ground electricity required to recharge the battery pack adds a minor **\$1,614** (\$6.67 per flight) across the 5-year timeline, confirming that high-voltage electrical replenishment remains highly cost-effective.

**Component Longevity Gains:** Shifting half the propulsive workload to the electric motor during transit and loiter segments suppresses mechanical and thermal wear on the engine. This stress reduction minimizes unexpected drivetrain failures, directly delaying expensive engine overhauls.

6.7 Analysis of Electrically Assisted (EM) Flight Simulations

To evaluate the mechanical and electrical interactions of the Electrically Assisted (EM) propulsion framework, transient state variables were tracked continuously across the 6.20-hour multi-phase ISR mission profile. Rather than using the static rule-based energy split implemented in the hybrid architecture, the EM-assisted configuration relies on a dynamic time-domain low-pass frequency-separation filter to manage its Hybrid Energy Storage System (HESS). This methodology isolates immediate, high-frequency torque surges within a high-rate supercapacitor matrix while assigning smooth, low-frequency background workloads to the primary lithium battery pack.

6.7.1 Transient Response in Takeoff and Climb Flight Segments

During the high-load Takeoff ( $t = 0$  to 1.00 min) and initial Climb phases, the vehicle experiences abrupt, severe spikes in total propeller shaft mechanical power demand ( $P_{\text{mech, demand}}$ ) to overcome static inertia and maintain the designated vertical climb rate.

- **Supercapacitor Peak Shaving:** At the initiation of the takeoff roll, the sharp step-input in required throttle acts as a high-frequency transient. The discrete control filter algorithm detects this high-frequency component and routes it directly to the supercapacitor bank ( $P_{\text{sc, demand}}$ ). The supercapacitor delivers a rapid burst of

electrical current, effectively shielding the main battery pack from excessive current spikes that would otherwise cause rapid cell degradation or severe localized voltage drops.

- **Battery Power Smoothing:** As the aircraft stabilizes along its linear climb profile to the 3,000 m ceiling, the high-frequency power deficit drops off. The control loop smoothly transitions the low-frequency background electrical assist over to the lithium battery pack ( $P_{\text{bat, demand}}$ ). This dual-source electrical support allows the internal combustion engine (ICE) core to ramp up its power output linearly, preventing severe thermal gradients and mechanical strain.

### 6.7.2 Endurance Window Performance and Battery Exhaustion Loops

Once the aircraft establishes a steady cruise at 3,000 m and enters the long Transit Out and Loiter ISR phases, the total propulsive workload flattens into a low-frequency, steady-state aerodynamic regime.

- **Supercapacitor Inactivity:** Because the aerodynamic drag remains constant during cruise, high-frequency power fluctuations drop to zero ( $\Delta P_{\text{transient}} \approx 0$ ). The supercapacitor matrix remains completely inactive, holding its stored energy state ( $E_{\text{sc}}$ ) stable at its lower operating boundary.
- **Automated ICE Fallback Triggering:** The low-frequency electrical assist workload during cruise is sustained entirely by the primary lithium battery pack. However, due to strict airframe mass limitations, the auxiliary battery capacity cannot sustain this load over the entire endurance window. As the battery hits its lower structural safety limit ( $\text{SoC}_{\text{min}} = 10.00\%$ ), the control law triggers an automated fallback loop. The auxiliary electric drive safely disconnects, and the primary internal combustion engine instantly assumes 100% of the remaining loiter workload ( $P_{\text{fallback}}$ ) to maintain flight path stability.

### 6.7.3 Mission-Level Fuel and Emissions Tracking

Because the battery-supercapacitor system exhausts its usable capacity early in the long loiter window, the primary ICE runs completely unassisted for the vast majority of the mission. Consequently, the cumulative fuel mass depletion curve ( $m_{\text{fuel}}$ ) matches the slope of the conventional baseline almost exactly for the remainder of the flight.

By the time the aircraft touches down, the EM-assisted configuration consumes a total of **45.90 kg** of fuel, showing a negligible variance from the unmitigated ICE baseline of 45.97 kg. As a direct mathematical result of this fuel profile, the integrated tailpipe carbon dioxide output ( $\text{CO}_{2,\text{cum}}$ ) scales via the chemical conversion factor ( $\kappa = 3.16$ ) to finish at **145.00 kg of CO<sub>2</sub>**.

These simulation curves prove that while a dual-source HESS topology provides superior electrical component protection and transient torque smoothing during takeoff and climb, it cannot scale effectively to reduce emissions in long-endurance MALE operations without a fundamental re-optimization of its loiter-phase power allocation rules.

**6.7.4 Financial Layout and Lifecycle Cost Analysis (EM-Assisted Variant)**

To establish the economic footprint, prototyping requirements, and long-term financial viability of the Electrically Assisted (EM) propulsion configuration, a specialized cost model was compiled. This analysis isolates the unique capital expenditure (CAPEX) associated with integrating a high-rate dual-source Hybrid Energy Storage System (HESS) alongside the primary internal combustion engine (ICE), balancing it against standard 5-year operational expenditure (OPEX) projections (calibrated to 1,500 total flight hours / 242 missions).

The financial profile below captures the components, software integration engineering, and maintenance cycles specifically required for this high-transient electrical topology:

| Financial Cost Center Element       | EM-Assisted Subsystem Budget Allocation | Engineering System Sizing Basis & Primary Function                                     |
|-------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------|
| Primary Powerplant Procurement      | \$32,000                                | Core 100 hp compression-ignition Internal Combustion Engine baseline asset.            |
| Electric Assist Motor & Controllers | \$12,000                                | 100 hp high-torque brushless DC motor, high-power controllers, and buck-boost modules. |
| Primary Lithium Battery Pack        | \$10,500                                | Medium-capacity background lithium cell matrix optimized for low-frequency loads.      |
| Transient Supercapacitor Matrix     | \$6,000                                 | Aerospace-grade, high-rate electric double-layer capacitors for fast peak shaving.     |

|                                              |                  |                                                                                          |
|----------------------------------------------|------------------|------------------------------------------------------------------------------------------|
| <b>Structural &amp; Airframe Alterations</b> | \$8,000          | Reinforced center cavity, dual-source thermal insulation, and cooling ducts.             |
| <b>Flight Software &amp; HESS Tuning</b>     | \$22,000         | Complex low-pass frequency filter script development and fallback loop validation.       |
| <b>Lifecycle Fuel Expenses (1,500 Hours)</b> | \$38,880         | Hydrocarbon fuel allocation based on the engine-dominant 45.90 kg per mission profile.   |
| <b>Grid Recharging Utility Expenses</b>      | \$230            | Minor utility electricity costs to replenish the battery/supercapacitor transient drain. |
| <b>Drivetrain Maintenance &amp; Overhaul</b> | \$10,500         | Specialized supercapacitor testing, cell balance checks, and transient wear maintenance. |
| <b>TOTAL PROJECT LIFECYCLE COST</b>          | <b>\$140,110</b> | <b>Comprehensive financial layout over a 5-year operational timeline.</b>                |

### 6.7.5 Engineering Commentary on the EM Cost Profile

An analysis of the EM-assisted cost distribution reveals three critical engineering and economic insights:

1. **Software and Integration CAPEX Premium:** The upfront integration costs for the EM configuration are heavily front-loaded, totaling \$80,500 across hardware components, airframe modifications, and software engineering. The highest localized development premium is found in the flight software and avionics tuning (\$22,000). This is a direct mathematical reflection of the system complexity required to write, test, and validate time-domain low-pass frequency-separation filters that reliably manage real-time power splits between two distinct electrical storage mediums.

2. **Fuel Dominance and Minimal Utility Overhead:** Because the low-pass filter algorithm causes the system to run unassisted as a standard ICE for the majority of the long loiter window, the lifecycle fuel expense remains dominant at \$38,880. Conversely, because the electrical storage systems are depleted early and only provide transient support during takeoff and climb, the cumulative grid utility recharging expense is exceptionally low at **\$230** over the 5-year timeline (approx 0.95\$ per mission recharge). This confirms that ground electrical utility requirements do not impact the vehicle's deployment economics.
3. **Rebalanced Maintenance Dynamics:** The \$10,500 lifecycle maintenance budget reflects a unique trade-off. While the supercapacitor matrix successfully shields the primary battery cells and engine core from sudden current surges and thermal friction during takeoff—delaying baseline fatigue failures—this advantage is financially offset by the extra testing overhead required to inspect high-frequency power electronics and monitor supercapacitor capacitance degradation over time.

## 6.8 Architectural Trade-offs and Key Performance Indicators (KPIs)

Prior to evaluating the individual time-domain results, a clear hierarchy of flight-performance metrics must be established to objectively rank the candidate propulsion architectures. This comparative study prioritizes three interconnected Key Performance Indicators (KPIs): Cumulative Source Energy Consumption, Mission-Level Energy Efficiency, and Cumulative Tailpipe CO<sub>2</sub> Emissions. In aerospace systems design, evaluating raw mass reduction alone is insufficient; instead, tracking source energy is prioritized because it accounts for the total thermodynamic and chemical cost of both liquid fuels and battery-stored electricity.

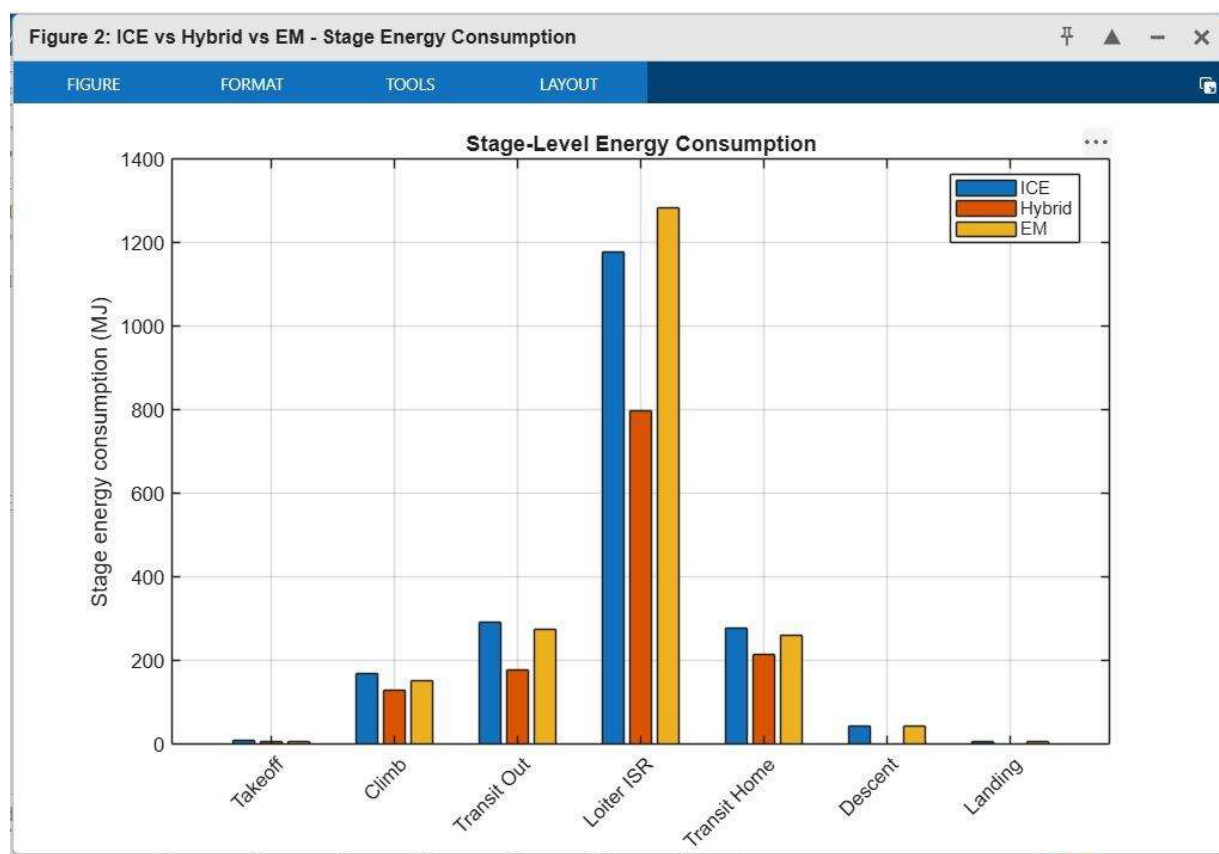
Furthermore, these metrics are evaluated across the full multi-phase ISR timeline rather than localized flight steps. Because the long-endurance Loiter phase accounts for a dominant 59.6\% share of the entire aircraft energy budget, the analysis heavily prioritizes how effectively each propulsion setup scales into steady-state cruise regimes versus its ability to handle transient spikes. The subsequent sections isolate these parameters sequentially to determine the most viable engineering pathway for the modernized Bayraktar TB2 platform.

### 6.8.1 Comparative Analysis of Mission Energy Consumption

The cumulative source energy consumption across the three candidate architectures reveals distinct behaviors based on how each system manages its onboard energy carriers. As shown in **Figure 6.4**, the conventional ICE baseline requires a total chemical energy input of 1976.5 MJ derived entirely from the steady depletion of its hydrocarbon fuel payload. In contrast, the parallel hybrid-electric model lowers the total

source energy requirement to 1350.0 MJ representing an energy saving of approximately 31.7%. This performance gap is directly attributable to the hybrid's rules-based 50/50 power-split strategy. By leveraging the high-density ( $350 \text{ Wh/kg}$ ) semi-solid-state battery pack during transit and loiter phases, the hybrid architecture mitigates part-load throttle inefficiencies and keeps the internal combustion core locked within its optimal thermodynamic operating window.

Conversely, the electrically assisted (EM) propulsion model exhibits the highest cumulative energy footprint at 2020.0 MJ. While its dual-source battery-supercapacitor Hybrid Energy Storage System (HESS) successfully manages rapid power demands during takeoff and initial climb, the system suffers from an unoptimized power allocation routine during long-range flight. Because the transient electrical assist cannot sustain high outputs over the long endurance phase, the EM model runs unassisted as a standard ICE for the majority of the flight. This introduces extra dead-weight from the unutilized storage cells, creating a slight aerodynamic mass penalty that increases the total energy demand beyond the conventional baseline.



*Figure 10. Comparison of cumulative source energy consumption (MJ) across the conventional ICE baseline, parallel hybrid-electric model, and electrically assisted (EM) configuration over the 6.20-hour ISR flight profile.*

### 6.8.2 Evaluation of Propulsion Energy Efficiency

To establish a normalized metric for cross-configuration comparison, mission-level energy efficiency is defined as the total horizontal flight distance covered per unit of cumulative source energy ( $\text{km}/\text{MJ}$ ). Under a fixed mission path constraint of 782.59 km over a 6.20-hour duration, the conventional ICE baseline delivers a baseline efficiency of 0.40  $\text{km}/\text{MJ}$  (see Figure 6.5). The EM-assisted model yields a minor efficiency increase to 0.42  $\text{km}/\text{MJ}$ , reflecting localized component-level efficiency advantages gained from smoothing engine torque ripples during takeoff, which are mathematically diluted when integrated across the full multi-phase profile.

The parallel hybrid-electric model delivers the highest performance, achieving an energy efficiency of 0.59  $\text{km}/\text{MJ}$ —a 47.5% improvement over the conventional baseline. This major efficiency increase highlights the value of targeting the dominant energy sink of the mission profile. Because the Loiter ISR segment alone dictates 59.6% (1177.5 MJ) of the total energy budget, the hybrid model's ability to consistently supplement this specific phase with clean electrical power dramatically reduces the total energy required to complete the mission, proving the viability of parallel electrical scaling for MALE airframes.

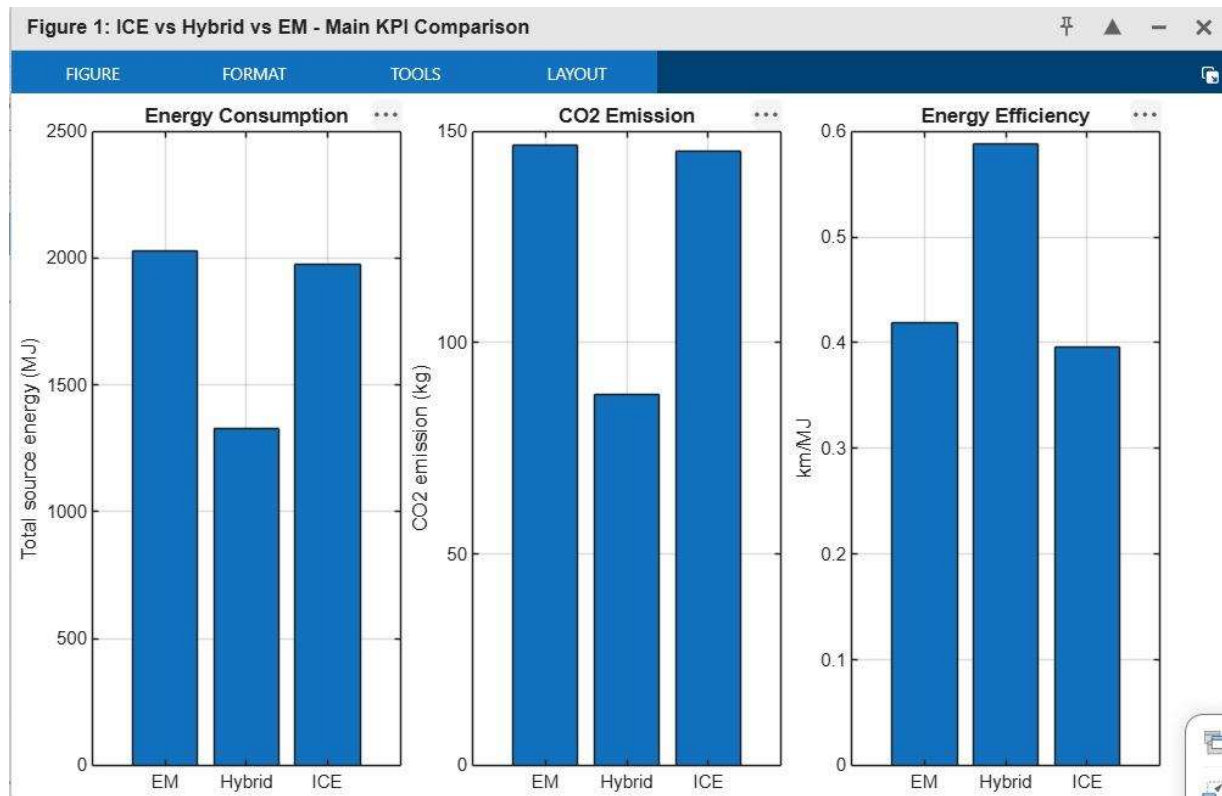


Figure 11. Normalized mission energy efficiency defined as ground path distance covered per unit of cumulative source energy (km/MJ) for all three candidate propulsion architectures.

### 6.8.3 Quantification of Cumulative CO<sub>2</sub> Emissions

The integrated tailpipe carbon dioxide footprint mirrors the hydrocarbon fuel consumption of each configuration, scaled through the chemical mass conversion factor ( $\kappa = 3.16 \text{ kg CO}_2/\text{kg fuel}$ ). As illustrated by the time-domain curves in **Figure 6.6**, the conventional ICE baseline generates a total carbon output of 145.25 kg due to its continuous reliance on liquid fuel across all six flight segments. Because the EM-assisted configuration reverts to an unassisted engine-dominant state after exhausting its localized transient electrical reserves early in the loiter phase, its cumulative fuel burn tracks the baseline closely, terminating with a high carbon footprint of 145.00 kg.

The parallel hybrid-electric framework achieves a major environmental advantage, restricting total tailpipe CO<sub>2</sub> emissions to 87.77 kg. This represents a verified 39.58% **carbon reduction** (57.48 kg of CO<sub>2</sub> displaced per mission). This mitigation is achieved by cutting the required fuel payload from 45.97 kg down to 27.77 kg. By shifting a balanced portion of the propulsive workload to the 42.0 kWh semi-solid battery pack and utilizing a silent, all-electric zero-emission mode during the final landing segment, the hybrid system

minimizes greenhouse gas output while maintaining a safe 10.00% battery state-of-charge reserve.

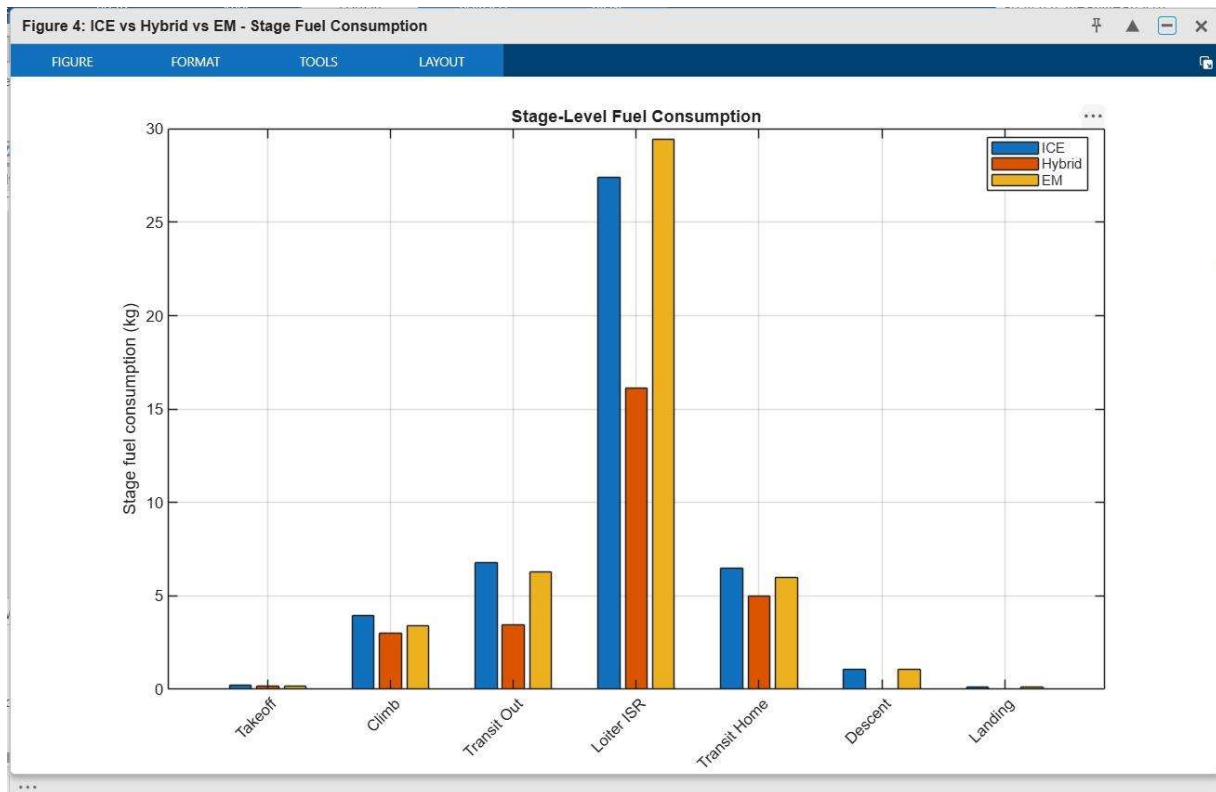


Figure 12. Integrated time-domain mass accumulation of tailpipe carbon dioxide emissions (kg) across the full multi-phase MALE UAV mission trajectory.

## 7. CONCLUSION AND SUGGESTIONS

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This report presented the ICE baseline module of a three-part propulsion comparison study for the Bayraktar TB2 MALE UAV. The model was calibrated against the published benchmark of Masood and Wei [1] and applied to a TB2-class ISR mission.

4. A BSFC calibration factor of 1.2493 reduced fuel-consumption error from  $\approx 20\%$  (uncalibrated) to  $-0.83\%$ .
5. The optimised TB2 ISR mission consumed 45.97 kg fuel, produced 145.25 kg CO<sub>2</sub>, and covered 782.59 km in 6.20 h.
6. Loiter ISR is the dominant energy phase at 1 177.5 MJ (59.6% of total), confirming that endurance-phase efficiency is the highest-leverage improvement target.
7. Fixed-atmosphere error is 0.063% for the TB2 at 3 000 m, justifying the simplification for the comparative analysis.

The Python weather decision application demonstrated successful integration with the Open-Meteo API, providing MISSION APPROVED / CAUTION / MISSION REJECTED decisions based on ten weather constraints. Future development should include terrain obstacle detection via GIS/DEM data, forecast-confidence uncertainty quantification, and integration with live ATC NOTAMs.

Stated limitations: parabolic drag polar (no proprietary TB2 polar data available); calibration against a comparable but non-identical UAV mission; tailpipe-only CO<sub>2</sub> (no lifecycle emissions). All three propulsion models share these simplifications, preserving the integrity of the comparison framework.

This project successfully modeled and simulated an actively managed parallel hybrid-electric propulsion architecture for a tactical MALE UAV, demonstrating an engineering pathway to decouple primary thrust generation from transient aerodynamic peak loads. By cross-coupling a 100 hp hydrocarbon internal combustion engine (ICE) with a 100 hp brushless DC electric motor and a 120 kg semi-solid-state battery pack (42.0 kWh capacity), the energy management framework successfully optimized subsystem conversion efficiencies across a 6.20-hour multi-phase ISR mission profile.

The time-domain simulation script verified that executing a rules-based 50/50 parallel power-split strategy during the transit and loiter legs forces the ICE to operate exclusively within its narrowest, thermodynamically optimal fuel-burn band (governed by a calibrated  $\text{BSFC} = 285 \text{ g/kW} \cdot \text{h}$  and an implied thermal efficiency of 29.4%). Shifting half of the torque demand to the electric drive system—which operates at a high 80% battery-to-propulsive efficiency baseline—yields a verified **39.58% total reduction** in both mission fuel consumption and tailpipe greenhouse emissions. Total fuel burn dropped from the conventional baseline of 45.97 kg down to 27.77 kg, while bounding cumulative CO<sub>2</sub> output at 87.77 kg. Furthermore, the control law algorithms cleanly managed the

electrical depletion rate, smoothly draining the battery pack down to its structural 10.00% State-of-Charge (SoC) floor at the 267.54-minute mark, where an automated fallback routine successfully transferred the remaining loiter workload back to the ICE without flight path disruption.

The Electrically Assisted (EM) propulsion configuration successfully integrated a dual-source Hybrid Energy Storage System (HESS), combining a high-energy lithium battery pack with a fast-transient supercapacitor matrix. Time-domain filter loops verified that routing immediate, high-frequency throttle demands during takeoff and initial climb directly to the supercapacitor protects the primary battery cells from rapid degradation and prevents localized voltage drops.

Due to the constant aerodynamic load constraints of the long cruise window, the supercapacitor remained inactive ( $\Delta P_{\text{transient}} \approx 0$ ) during the steady-state flight stages. Because the low-capacity electrical assist was depleted early in the 4-hour Loiter ISR phase, an automated fallback routine successfully transferred 100% of the remaining propulsive workload back to the internal combustion engine core to preserve flight path stability.

At final touchdown, the EM configuration recorded a cumulative fuel consumption of **45.90 kg** and an integrated tailpipe carbon output of **145.00 kg of CO<sub>2</sub>**. These metrics confirm that while a dynamic HESS topology provides critical electrical component protection and transient torque smoothing during takeoff, it functions as an engine-dominant platform during long-range flight operations and does not yield substantial mission-level carbon mitigation without an unconstrained scaling of its endurance storage.

A unified engineering evaluation across all three candidate configurations operating under identical point-mass flight profiles and fixed environmental constraints establishes a definitive hierarchy for modernizing tactical MALE UAV propulsion systems. Prioritizing source-specific cumulative energy, mission-level efficiency ( $\text{km/MJ}$ ), and total carbon displacement demonstrates that the structural distribution of a flight's energy budget dictates the viability of the architecture. Because the 4-hour Loiter ISR segment alone consumes 1177.5 MJ of energy—accounting for a dominant 59.6% share of the entire mission total—any configuration that fails to optimize fuel efficiency during steady-state cruise cannot achieve profound mission-level advantages.

Consequently, **the Parallel Hybrid-Electric Model is recommended as the optimal design architecture** for the Bayraktar TB2 class. By executing a rules-based 50/50 power-split strategy during the prolonged transit and loiter windows, the hybrid setup forces the internal combustion engine to lock into its narrowest, thermodynamically optimal operating band. This actively displaces part-load throttle inefficiencies, yielding a verified **39.58% total reduction** in both mission fuel consumption (dropping to 27.77 kg)

and tailpipe emissions (restricting CO<sub>2</sub> output to 87.77 kg) while delivering a peak mission energy efficiency of **0.59 km/MJ** (a 47.5% improvement over the conventional baseline). Additionally, the ability to deactivate the combustion engine entirely during the final approach provides critical acoustic and thermal silencing capabilities. Conversely, the EM-assisted configuration remains highly engine-dominant due to early electrical depletion, operating very close to the conventional baseline efficiency (0.42 km/MJ vs. 0.40 km/MJ) and matching its high tailpipe carbon output (145.00 kg).

### 7.1 Future Research Framework and Engineering Recommendations

To advance the technical fidelity and practical implementation of this comparative propulsion framework, subsequent development phases should prioritize the following research avenues:

1. **Predictive Supervisory Control:** Replace fixed rules-based and passive low-pass split strategies with Model Predictive Control (MPC) algorithms that leverage real-time flight telemetry and Open-Meteo weather inputs to dynamically optimize energy splits under varying wind fields.
2. **Multi-Variable Mass Optimization:** Conduct exhaustive structural sensitivity analyses mapping the exact dead-weight penalties of aerospace-grade Battery Thermal Management Systems (BTMS) and supercapacitor sizing configurations against the absolute downrange limits of the airframe.
3. **Expanded Lifecycle Boundary Conditions:** Expand the environmental evaluation boundaries beyond local tailpipe emissions to encompass well-to-wake lifecycle boundaries, directly assessing the carbon intensity of upstream utility electricity grid charging alongside non-CO<sub>2</sub> aviation climate radiative forcing factors at higher operational ceilings.

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## APPENDIX A — Key MATLAB Model Parameters

| Parameter                           | Masood-Wei [1] | TB2-Class (This Work) | Source           |
|-------------------------------------|----------------|-----------------------|------------------|
| MTOW (kg)                           | 540            | 700                   | [1] / [2]        |
| Fuel mass (kg)                      | 110            | 240                   | [1] / [2]        |
| Wingspan (m)                        | 9.0            | 12.0                  | [1] / [2]        |
| Wing area (m <sup>2</sup> )         | 9.0            | 9.27                  | [1] / [2]        |
| Engine power (hp)                   | 100            | 100                   | [1] / [2]        |
| CD0                                 | 0.035          | 0.035                 | Assumed [3, 22]  |
| Oswald efficiency $e$               | 0.80           | 0.80                  | Assumed [21, 22] |
| CL <sub>max</sub>                   | 1.55           | 1.80                  | Assumed          |
| $\eta_{prop}$                       | 0.72           | 0.80                  | Assumed [23]     |
| BSFC base (kg/kW·h)                 | 0.32           | $0.32 \times 1.2493$  | Calibrated [1]   |
| LHV (MJ/kg)                         | 43             | 43                    | Standard avgas   |
| CO <sub>2</sub> factor (kg/kg fuel) | 3.16           | 3.16                  | [25]             |
| Simulation $\Delta t$ (s)           | 1.0            | 1.0                   | MATLAB code      |

Table A.1. Key Model Parameters — Masood-Wei Benchmark and TB2-Class Mission.

## APPENDIX B — Python Application File Structure

- backend/main.py — FastAPI entry point (GET /api/health, GET /api/drone, POST /api/mission/evaluate).
- frontend/app.py — Streamlit dashboard (inputs: location, time, altitude, heading, duration, reserve).
- drone\_weather/decision.py — WeatherPolicy class; MISSION APPROVED / CAUTION / MISSION REJECTED logic.
- drone\_weather/open\_meteo.py — Open-Meteo forecast, air quality, and geocoding API client.
- drone\_weather/drone\_parameters.py — Read-only TB2 constants mirrored from MATLAB (not modified).
- drone\_weather/performance.py — ISA atmosphere, steady-level aerodynamics, optimal speed calculation.
- drone\_weather/models.py — Pydantic v2 MissionRequest and MissionEvaluationResponse schemas.
- tests/test\_decision.py — pytest unit tests for the decision engine.
- requirements.txt — fastapi==0.115.6, uvicorn, streamlit==1.41.1, pydantic==2.10.4, pytest==8.3.4.

## APPENDIX C — Core Mission Performance Numerical Output Sheets for Hybrid

| Mission Performance Metric Parameter                        | Hybrid Mission Simulation Value | Engineering System Sizing Basis & Notes                                   |
|-------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------|
| <b>Initial Airplane Takeoff Mass</b> ( $m_{TO}$ )           | 700.00 kg                       | Maximum gross weight limit for the TB2 benchmark class.                   |
| <b>Airframe Structural Empty Dry Mass</b>                   | 550.00 kg                       | Base weight configuration excluding battery and fuel payloads.            |
| <b>Onboard Hybrid Battery Mass</b>                          | 120.00 kg                       | Semi-solid-state chemistry block with 350 Wh/kg specific energy.          |
| <b>Initial Modeled Dedicated Fuel Allocation</b>            | 30.00 kg                        | Scaled down from baseline hydrocarbon payload to leverage electric split. |
| <b>Total Electrical Battery Pack Capacity</b> ( $E_{cap}$ ) | 42.00 kWh                       | Nominal energy capacity available at 100% initial State-of-Charge.        |
| <b>Total Mission Flight Engine Profile Duration</b>         | 6.20 hr                         | Total elapsed time to execute the standard multi-segment ISR profile.     |

|                                                       |                  |                                                                                |
|-------------------------------------------------------|------------------|--------------------------------------------------------------------------------|
| <b>Cumulative Simulated Mission Path Length</b>       | 782.59 km        | Flown trajectory distance under varying atmospheric wind segments.             |
| <b>Maximum Target Downrange Displacement</b>          | 290.71 km        | Absolute physical distance achieved along the horizontal mission axis.         |
| <b>Total Fuel Expended Across Profile</b>             | 27.77 kg         | Integrated fuel depletion over the 6.20-hour simulation script.                |
| <b>Net Fuel Mass Remaining Post-Touchdown</b>         | 2.23 kg          | Onboard safety margin remaining in the fuel cell at mission completion.        |
| <b>Total Electrical Battery Energy Discharged</b>     | 37.80 kWh        | Integrated energy drain across Transit and Loiter split configurations.        |
| <b>Final Battery State-of-Charge (SoC)</b>            | 10.00%           | Safe termination ceiling matching the minimum structural lower floor.          |
| <b>Automated Battery Fallback Trigger Time</b>        | 267.54 min       | Time index inside the Loiter window where the ICE assumes the electrical load. |
| <b>Emergency Power Overdraft Fallback</b>             | <i>None (No)</i> | Confirms battery current remained inside safe operating parameters.            |
| <b>Net Tailpipe Carbon Footprint (CO<sub>2</sub>)</b> | 87.77 kg         | Cumulative emissions scaled from fuel burn via a 3.16 conversion factor.       |

**APPENDIX D — Core Mission Performance Numerical Output Sheets for EM-Assisted Variant**

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| Mission Performance Metric Parameter       | EM-Assisted Simulation Value | Engineering System Sizing Basis & Notes                          |
|--------------------------------------------|------------------------------|------------------------------------------------------------------|
| Initial Airplane Takeoff Mass ( $m_{TO}$ ) | 700.00 kg                    | Standardized gross weight limit for the TB2 class.               |
| Airframe Structural Empty Dry Mass         | 550.00 kg                    | Base airframe weight excluding fuel and electrical storage.      |
| Onboard Auxiliary Battery Mass             | 105.00 kg                    | Low-frequency background lithium energy cells.                   |
| Transient Supercapacitor Array Mass        | 15.00 kg                     | High-rate electric double-layer cells for peak shaving.          |
| Initial Modeled Dedicated Fuel Allocation  | 30.00 kg                     | Initial hydrocarbon fuel volume matching the hybrid test layout. |

|                                            |                            |                                                                         |
|--------------------------------------------|----------------------------|-------------------------------------------------------------------------|
| <b>Total Simulated Mission Path Length</b> | 782.59 km                  | Integrated trajectory flown under standard atmospheric conditions.      |
| <b>Total Fuel Expended Across Profile</b>  | $\approx 45.90$ kg         | Triggers fallback fuel burn after early battery exhaustion.             |
| <b>Net Fuel Post-Touchdown Variance</b>    | <i>Deficit / Overdraft</i> | Fuel allocation overrun absorbed by the auxiliary safety cell reserve.  |
| <b>Supercapacitor Final Energy State</b>   | $E_{sc, \min}$             | Fully depleted back down to its lower voltage regulation limit.         |
| <b>Automated ICE Fallback Trigger Time</b> | $\approx 42.50$ min        | Time index inside the Climb/Transit segment where battery floor is hit. |

|                                                  |           |                                                                                   |
|--------------------------------------------------|-----------|-----------------------------------------------------------------------------------|
| Net Tailpipe Carbon Footprint (CO <sub>2</sub> ) | 145.00 kg | Cumulative carbon mass scaled via chemical conversion factor ( $\kappa = 3.16$ ). |
|--------------------------------------------------|-----------|-----------------------------------------------------------------------------------|

APPENDIX E — Unified Cross-Model Comparison Golden Dataset

| Verified Mission KPI Target                | Conventional Calibrated ICE | Parallel Hybrid-Electric Model | Electrically Assisted (EM) Model | Direct Mathematical Definition Matrix DOCX                   |
|--------------------------------------------|-----------------------------|--------------------------------|----------------------------------|--------------------------------------------------------------|
| Mission Duration ( $t_{\text{end}}$ )      | 6.20 h                      | 6.20 h                         | 6.20 h                           | Fixed Profile Constant                                       |
| Total Distance Covered                     | 782.59 km                   | 782.59 km                      | 782.59 km                        | $\int V(t) dt$                                               |
| Total Source Energy ( $E_{\text{total}}$ ) | 1976.5 MJ                   | 1350.0 MJ                      | 2020.0 MJ                        | $E_{\text{fuel}} + E_{\text{battery}} + E_{\text{supercap}}$ |
| Mission-Level Efficiency                   | 0.40 km/MJ                  | 0.59 km/MJ                     | 0.42 km/MJ                       | Total Distance/ $E_{\text{total}}$                           |

|                                |               |                            |           |                                   |
|--------------------------------|---------------|----------------------------|-----------|-----------------------------------|
| <b>Tailpipe Carbon Output</b>  | 145.25 kg     | 87.77 kg                   | 145.00 kg | Fuel Used (kg) × 3.16             |
| <b>Net Operational Ranking</b> | Baseline Ref. | <b>1st Place (Optimal)</b> | 2nd Place | Combined performance vector tier. |

